Back into Plato's Cave: Examining Cross-modal Representational Convergence at Scale

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Abstract

The Platonic Representation Hypothesis (Huh et al., 2024) posits that neural networks trained on different modalities (e.g., text and images) converge toward a shared representation of reality. If true, this has significant implications for whether modality choice matters at all. In this paper, we show that the evidence for this claim is substantially weaker than subsequent work suggests. The mutual k-nearest-neighbor metric used on 1024 text–image pairs in the original study captures only coarse structure. To keep the alignment from collapsing as one scales up the data, k has to grow proportionally, undercutting the argument for fine-grained representational convergence. The reported increase in alignment with language model strength saturates for recent models. Moreover, the one-to-one text–image pairing favors alignment, while alignment decreases with non-bijective data. We further find that image and text representations indeed share coarse semantic structure, but neither stronger language models nor richer captions yield fine-grained alignment. Thus, multimodal representations share coarse structure without evidence of convergence to a shared representation – arguably, full representational convergence would require fine-grained alignment.

Project page: `https://akoepke.github.io/cave_umwelten`

1 Introduction

The success of Large Language Models (LLMs) is causing hand-wringing in the computer vision community: do we need pixels to build machines that understand our world, or is language "all you need"?

Several works have demonstrated that models trained only on text data successfully solve what were thought to be fundamentally visual problems, such as visual question answering (Gu et al., 2023; Hu et al., 2023b), visual reasoning (Chollet, 2019; Hu et al., 2023a; Wang et al., 2023b; Akyürek et al., 2024), or embodied robotics applications (Ahn et al., 2022; Liang et al., 2023). This resonates with the suggestion that text data may make other modalities redundant (Sutskever, 2023), on the premise that the part of the world that is relevant to humans is manifest in language. On the other hand, it is argued that linguistic data alone cannot yield genuine understanding (Browning & LeCun, 2022) or actual embodiment. After all, there is a reason we visit art museums rather than just read descriptions of paintings in a catalogue. This raises a central question: how do models trained on different modalities represent reality?

The Platonic Representation Hypothesis (Huh et al., 2024) offers a compelling answer: it suggests that as neural networks grow larger and consume more data, their learned representations will eventually converge toward a common representation of the underlying world, no matter which data modality (text, vision, audio, touch, etc.) was used for training. Proponents of language-only learning have interpreted this as validation of their approach. Since, according to the Platonic Representation Hypothesis, the choice of modality does not matter, as they all lead to the same shared representation, one might as well use language as the most convenient source of data.[1] However, a hypothesis is only as strong as its evidence. Here, the evidence rests on methodological choices that have largely gone unexamined.

[1]But analogously, the same argument could be made for vision-only learning (Isola, 2025).

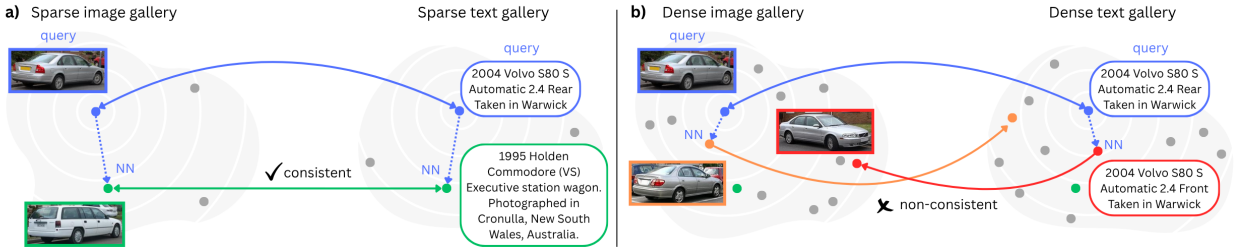


Figure 1: Illustration of the mutual nearest neighbor metric used by Huh et al. (2024) to measure cross-modal alignment. (a) Sparse regime: given a query image and caption (blue), nearest neighbors (NN) are retrieved independently in image and text embedding spaces. Mutual NN alignment measures whether the NNs are consistent across modalities. (b) Dense regime: as the dataset size increases, NNs within each modality get better. The vision model retrieves a car in the same pose, and the language model retrieves a caption of the same car model regardless of pose. Within-modality organization does not necessarily agree across modalities.

In the first part of this paper, we take a close look at the experimental setting of Huh et al. (2024). They conducted their analysis on small, sparse datasets with one-to-one correspondences between modalities. However, real-world multi-modal data is large, dense, and inherently many-to-many. One image has many valid descriptions, and a single caption can correspond to many plausible images. These differences change what it means for two representations to "align" under the mutual nearest neighbor metrics.

In a sparse dataset, models can appear aligned simply because no better alternatives exist (Fig. 1a). Here, two models can agree despite organizing their representations differently. As the dataset grows (i.e. the gallery used for retrieving neighbors gets denser), both models find closer neighbors (Fig. 1b). A vision model may retrieve an image of a car in the same pose as the query, while the language model retrieves a caption describing the query's car model. Both are individually valid, but produce a mismatch across modalities that the mutual nearest-neighbor metric penalizes. Mutual nearest-neighbor agreement with fixed k as used in Huh et al. (2024) is an increasingly fine-grained measure for alignment as the gallery grows.

Indeed, when we increase the gallery from 1024 to millions of samples, cross-modal mutual kNN agreement at fixed small k drops, while same-modality pairs remain stable under identical densification. Rerunning the original setup from Huh et al. (2024) with recent models and benchmarks, we further find that the reported alignment trend for increasingly strong language models saturates. Allowing multiple valid correspondences per sample reduces mutual kNN agreement, reflecting a limitation of the mutual kNN protocol.

In the second part of the paper, we characterize the cross-modal agreement that remains. We find that image and text models agree on what the world contains at a categorical level, while organizing fine-grained within-category structure differently. Neither stronger language models nor richer captions close this gap. The picture that emerges is less a shared Platonic reality than a set of overlapping Umwelten (Uexküll & Kriszat, 1934). Each modality agrees on the broad semantic categories yet perceives and arranges them in its own way. The evidence shows shared coarse semantics, not shared representations.

In this paper, we make the following observations:

Mutual kNN alignment for fixed small k degrades with scale: Increasing the gallery from 1024 to millions of samples causes a drop in cross-modal mutual nearest-neighbor agreement for fixed small neighborhood sizes, while agreement at $k{=}n/100$ remains stable (Section 3.2.1).

Previously reported trends do not continue under the original setup: The claim that stronger language models align better with vision weakens for more recent models and benchmarks (Section 3.2.3).

Many-to-many correspondence reduces mutual kNN alignment: Allowing multiple valid correspondences per sample leads to decay in mutual kNN agreement for fixed k (Section 3.2.4).

Modalities share coarse structure, not global organization: Cross-modal agreement on coarse structure approaches the same-modality ceiling. Global organization saturates well below it, and unimodal agreement stays high throughout (Section 4).

2 Related Work

One Platonic Ideal vs many *Umwelten*. In his "Theory of Forms", Plato (c. 375 BC) argued that every physical object we perceive is a flawed imitation (a shadow) of some eternal, abstract "ideal" form, and only by escaping from the tyranny of our physical senses (leaving the cave of shadows), we can achieve true understanding. But in the 20th century, this argument for a single, unified Platonic Ideal representation has been repeatedly undercut by biologists, psychologists, and philosophers. Uexküll & Kriszat (1934) argued that every organism inhabits its own perceptual environment, or *Umwelt*: a tick lives in a world of thermal gradients, a bat in a world of echoes. The different Umwelten might have only limited overlap with each other. [2]. Gibson (1979) pushed this further, proposing that perception is shaped by what an organism can *do* in its environment, not by an observer-independent reality. Wittgenstein, thinking about language, arrived at a strikingly similar conclusion. He famously argued: "If a lion could talk, we wouldn't be able to understand it" (Wittgenstein, 1953/2009), meaning that the lion's world (goals, instincts, perceived reality) is so utterly different from our own, that even if it spoke English, we would not comprehend the meaning. [3] Building on Wittgenstein, Rosch (1978) developed her Prototype Theory of Categorization, arguing against a single Platonic ideal as a representation of object categories, proposing a data-driven clustering-based model instead.

Representational alignment. The question of representational similarity has been studied extensively in the neurosciences (Edelman, 1998; Haxby et al., 2001; Kriegeskorte et al., 2008). In machine learning, the parallel question of whether independently trained networks learn similar internal structure has received growing attention. Lenc & Vedaldi (2015) investigated the equivalence of representations from different trained models and found that early convolutional layers are more interchangeable than later ones. This task, also referred to as "model stitching", was later revisited by Bansal et al. (2021). Related to this, Li et al. (2016) proposed methods to align neurons across independently trained networks. More recently, Dravid et al. (2023) introduced "Rosetta Neurons", showing that different vision models share common units corresponding to similar visual concepts across architectures, tasks, and training data. Tasker et al. (2026) propose the Universal Normal Embedding hypothesis, where encoder and generative latents are interpreted as noisy views of an approximately Gaussian shared space.

Furthermore, alignment has been linked with shared model capabilities measured by task performances (Balestriero et al., 2018; Jiang et al., 2024; Huh et al., 2024; Morcos et al., 2018; Bansal et al., 2021). To directly quantify representational similarities, several metrics have been used to measure correlations between features (Morcos et al., 2018; Hotelling, 1936). Kornblith et al. (2019) introduced Centered Kernel Alignment (CKA) as a robust measure invariant to orthogonal transformations and isotropic scaling. Huh et al. (2024) found the CKA metric to reveal only a "weak trend of alignment between models" and proposed the use of the mutual kNN metric that measures the overlap of two sets of neighborhoods of size k.

Multi-modal alignment. Early efforts to connect images and text utilized human annotations (Von Ahn et al., 2007). The curation of large-scale paired image-caption datasets, such as MS-COCO (Lin et al., 2014) and Visual Genome (Krishna et al., 2017), facilitated the systematic study of cross-modal correspondence and models. The CLIP model (Radford et al., 2021) formed a turning point by demonstrating that contrastive learning on web-scale image-text pairs could produce shared embedding spaces.

Since then, a growing body of work has investigated whether such alignment arises even without explicit joint training. Merullo et al. (2023) showed that a simple learned linear transformation could map between frozen vision encoders and LLMs. Moschella et al. (2023) use similarities to an anchor set. Maniparambil et al. (2024) demonstrated that even unaligned unimodal encoders possess high semantic similarity. Li et al. (2024) study alignment at the level of shared categories. Complementary to that, Lu et al. (2025b) define each sample by its angular distances to others, improving training-free alignment across vision, language, and audio.

Fully unsupervised approaches include blind vision-language matching (Schnaus et al., 2025) and unpaired embedding translation via cycle-consistency (Jha et al., 2025; Zhu et al., 2017). Finally, Gupta et al. (2026)

[2]For a tour of von Uexküll's ideas, see Koenderink's delightful book (Koenderink, 2019).

[3]For a great treatment of Wittgenstein's argument in popular culture, see the episode *Darmok* of the American TV series *Star Trek: The Next Generation*.

show that an orthogonal map can map between independently trained multi-modal contrastive models. These results are often seen as evidence for representational convergence, also supported by Lu et al. (2025a) in their survey of multi-modal alignment. However, those results are obtained in restricted settings (e.g. Schnaus et al. (2025) experiments on CIFAR-100 and ImageNet-100) and do not scale to real-world multi-modal data. Our work examines whether alignment survives beyond these constraints, showing that cross-modal alignment reflects coarse agreement rather than shared global structure.

Limits and measurement of emergent cross-modal structure. Several analyses show that alignment between independently trained unimodal encoders depends strongly on data, architecture, and evaluation protocol. Tjandrasuwita et al. (2025) find that alignment varies with modality similarity and the balance of shared versus unique information, while Hadgi et al. (2025) report weaker alignment for "pure" 3D encoders without careful subspace selection. Zhu et al. (2026) further show that video–text alignment depends on temporal information and text availability.

Gröger et al. (2026) show that measures such as CKA are sensitive to network scale and can be altered via null calibration, largely removing evidence of global convergence while leaving local neighborhood similarity (e.g. mutual kNN) stable, though their evaluation remains in small-scale regimes and on the original terse captions, which cap measurable alignment (Section B.2). Our granularity analysis in Section 3.2.1 suggests the calibration result and the local-persistence result are not two separate findings: at $n{\approx}1$k, mutual kNN with $k{=}10$ probes coarse categorical structure, so the "local" alignment that survives their calibration and the "global" alignment that vanishes are different views of the same coarse overlap. Beyond similarity metrics, Smith et al. (2025) and Kumar et al. (2025) show that functional agreement can persist even when internal representations are misaligned, suggesting that behavioral compatibility does not imply shared structure. These caveats echo arguments that text-only learning may be insufficient to recover perceptual structure (Bender & Koller, 2020; LeCun et al., 2022), and motivate multi-modal foundation models that integrate perception and language (Huang et al., 2023; Bai et al., 2025; Meta AI, 2025; Hong et al., 2025).

3 How much do representations align?

In this section, we take a close look at the experimental evidence underpinning the Platonic Representation Hypothesis (Huh et al., 2024). The experiments in Huh et al. (2024) rest on two foundations that warrant scrutiny: the use of mutual kNN alignment on a small evaluation set of only 1024 samples from the Wikipedia Image-Text (WIT) dataset (Srinivasan et al., 2021) (WIT-1024) with effectively bijective (one-to-one) image-text correspondences. This assumption is implicit in the WIT-1024 setup rather than stated explicitly in Huh et al. (2024). Typically, these choices are not acknowledged when the hypothesis is cited (Marcos-Manchón & Fuentemilla, 2026; Ruan et al., 2024; Liu et al., 2025; Chai et al., 2025; Dar, 2025). The hypothesis is usually invoked in its broad, appealing form rather than in the narrower terms under which experimental support was provided.

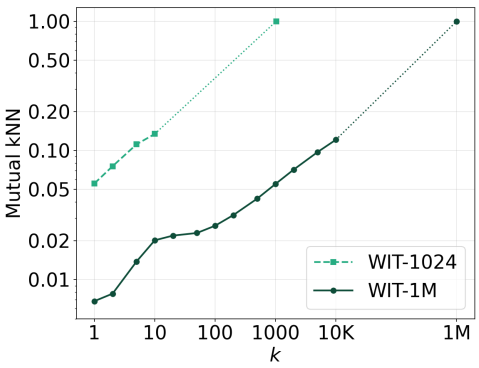
3.1 Experimental setup

Mutual kNN metric. To measure alignment between representations from different models, we follow Huh et al. (2024) in using the mutual k-nearest-neighbor metric (illustrated in Fig. 1). Given a shared gallery set of n datapoints (referred to as mini-batch sampled from the data distribution in Huh et al. (2024)) encoded into feature vectors $\mathbf{a_i} \in \mathbb{R}^{d_1}$ and $\mathbf{b_i} \in \mathbb{R}^{d_2}$ by two models with $i \in \{1, \cdots, n\}$, we first L2-normalize each representation. The features are then clamped to the 0.95-quantile of the norm of the activations (Hastings Jr et al., 1947). We then retrieve the k nearest neighbors of every query point independently for each model (e.g. image and text query for vision and language encoders):

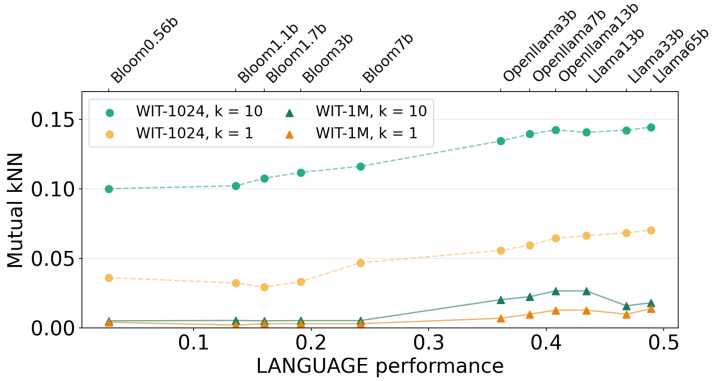
$$\mathcal{N}_k^{\mathbf{a}}(i) = \text{argtopk}_{j \neq i}\, \mathbf{a}_i^\top \mathbf{a}_j, \qquad \mathcal{N}_k^{\mathbf{b}}(i) = \text{argtopk}_{j \neq i}\, \mathbf{b}_i^\top \mathbf{b}_j.$$

The per-sample score is the number of overlapping samples normalized by k:

$$s_i = \frac{|\mathcal{N}_k^{\mathbf{a}}(i) \cap \mathcal{N}_k^{\mathbf{b}}(i)|}{k},$$



(a) Effect of neighborhood sizes k in mutual kNN (both axes log-scaled). Trivially, mutual kNN converges to 1 as k approaches the full gallery. Huh et al. (2024) utilized mutual kNN with $k = 10$.



(b) Alignment between DINOv2 and different LLMs, measured on WIT-1024 and WIT-1M. Huh et al. (2024) observed that alignment (mutual kNN) increases with language performance (measured as $1 -$ bitsperbyte on the 1024-sample set), but we find that this trend weakens with larger gallery size.

Figure 2: Mutual kNN text-image feature alignment when scaling from WIT-1024 to WIT-1M. (a) shows the dependence on neighborhood size k, while (b) examines alignment for different LLMs. *The observation from Huh et al. (2024), that more capable language models align better with vision at fixed small k is strongly weakened at WIT-1M scale.*

and the overall mutual-kNN score is the mean over all samples. A score of 1 means that every point's k nearest neighbors are identical in both spaces, and a score of 0 means that the k nearest neighbors do not overlap. In the sparse gallery in Fig. 1, the query (blue) retrieves the same neighbor in both image and text spaces, giving a mutual kNN score of 1 for k=1. A score of $\frac{k}{n}$ suggests chance-level expected overlap for independent random retrieval, which decreases for growing n. Chance-level overlap is $\approx 9.8\times10^{-3}$ for k=10 at n=1024, but only 10^{-5} at n=10^6. Correcting for chance, $\tilde{s} = (s - \frac{k}{n})/(1 - \frac{k}{n})$, changes scores by at most $\approx$0.009 at n=1024 and by less than 10^{-4} for $n \geq 10^5$, and does not affect any of the trends we report. Enrichment over chance ($\sim$800$\times$ at n=15M) is not the quantity at issue. Identical representations agree on neighborhoods at every granularity, so high enrichment with $\sim$1% absolute overlap signals partial coarse agreement, not convergence. Note that throughout this section, we report raw mutual kNN (as in Huh et al. (2024)) using WIT-1024 as the set of queries. We report results for different galleries: WIT-1024, WIT-1M, and LAION-15M.

Implementation details. For most experiments, we use DINOv2-base (Oquab et al., 2024) as the vision encoder. We refer to this model as DINOv2 in the following. Our primary language model is OpenLlama3b (Touvron et al., 2023; Geng & Liu, 2023), abbreviated as OpenLlama. For each image and text sample, we extract the representations from all layers of their respective encoders and follow the experimental protocol from Huh et al. (2024). Additional model pairs and the full model list used in Section 3 are provided in Section A.4 and Section G.3.2. We use Faiss (Douze et al., 2024) for nearest neighbor computation at scale, specifically `IndexFlatL2`, which is equivalent to using cosine similarity on normalized vectors.

3.2 Probing alignment as measured by mutual kNN

Here, we analyze how alignment as measured by mutual kNN behaves at different levels of granularity (k=1, k=10, k=$\frac{n}{100}$, and on a denser gallery (million(s of) instead of 1024 samples). We then turn to the bijective assumption and examine what happens when it is relaxed. We further test whether these patterns extend beyond text-image to the text-audio and text-video settings. Finally, we perform a trend check to ask whether the predictions from Huh et al. (2024) have held up as models have improved.

Sensitivity to k in mutual kNN. Huh et al. (2024) reported mutual kNN alignment for k=10. As shown in Fig. 2a, the metric trivially converges to 1 as k approaches the full gallery size n, since both neighbor sets then contain all samples. In our analyses at larger gallery scales, we perform deduplication to prevent near-duplicate samples from trivially inflating neighborhood overlap (see Section G.1 for details).

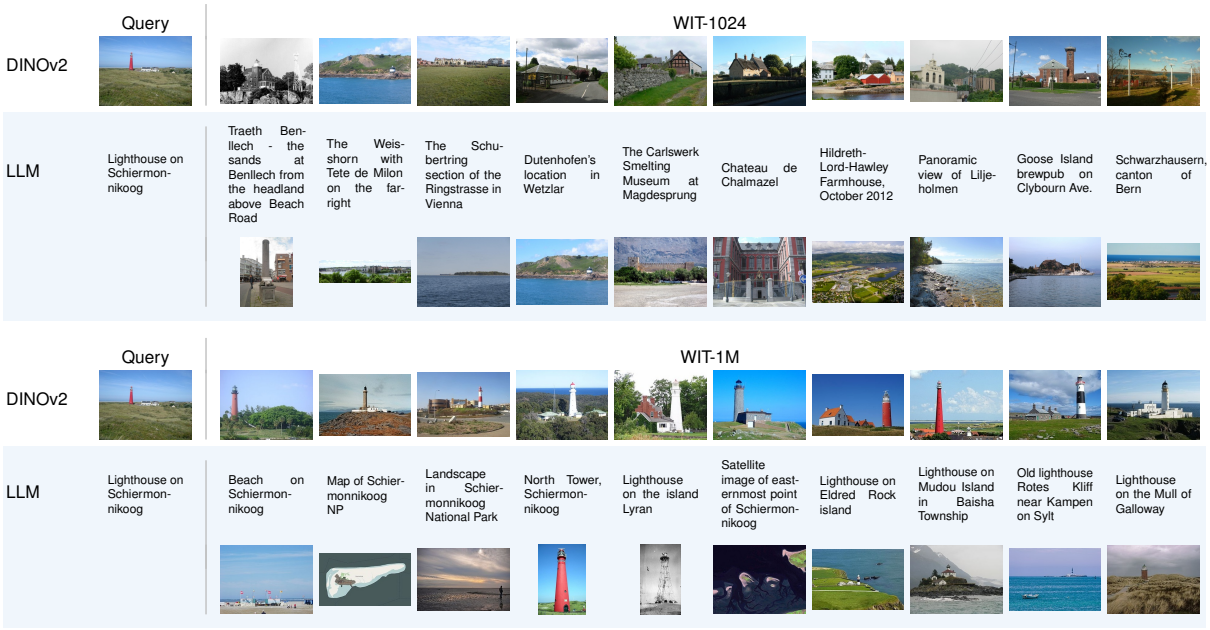


Figure 3: **Nearest-neighbor quality depends on data density.** We show 10 within-modality nearest neighbors for image (DINOv2) and text (LLM) embeddings on a sparse WIT-1024 gallery (top) and a denser WIT-1M gallery (bottom). For text queries, retrieved captions and their corresponding reference images are shown. At smaller scale, nearest neighbors are less semantically precise. Nearest-neighbor structure becomes more semantically refined as gallery density increases.

3.2.1 Alignment across dataset scales

Nearest neighbors in sparse gallery. We now turn to the data and ask whether the 1024-sample gallery used in Huh et al. (2024) is too sparse to capture more than coarse structural agreement. As shown in Table 1, the mean cosine similarity between queries and nearest neighbors in terms of both image (DINOv2) and text features (OpenLlama) is significantly lower for the WIT-1024 gallery compared to WIT-1M (e.g. 0.799 compared to 0.906 for DINOv2 at k=1). For both, we use WIT-1024 as the query set.

We visualize the 10 nearest neighbors for image and text features on WIT-1024 and WIT-1M in Fig. 3. At low density, semantically unrelated samples can appear as nearest neighbors as there is nothing closer available, meaning that measured mutual kNN agreement can reflect the shared lack of alternatives. To get more meaningful insights, we scale the density of the retrieval gallery in the following subsection.

Table 1: As the gallery grows, nearest neighbors get closer to the query set in both DINOv2 and OpenLlama embedding spaces, facilitating a more fine-grained analysis of cross-modal alignment.

Gallery	Model	k=1	k=10
WIT-1024	DINOv2	0.799	0.717
WIT-1024	OpenLlama	0.502	0.400
WIT-1M	DINOv2	0.906	0.888
WIT-1M	OpenLlama	0.757	0.701

Densification by scaling the gallery size. Having established that WIT-1024 captures mainly coarse structure, we test whether alignment persists as we densify the gallery up to 1M (English WIT (Srinivasan et al., 2021)) and 15M samples (LAION-400M (Schuhmann et al., 2021)). Best layer pairs are chosen following Huh et al. (2024) (see Section A.2).

As shown in Fig. 4, alignment scores decrease as gallery size n grows for fixed k. On WIT-1024, mutual kNN scores are 0.135 ($k = 10$) and 0.058 ($k = 1$), but drop to 0.008 and 0.001 on LAION-15M. As the neighborhood considered by the metric shrinks by four orders of magnitude, these fixed-k sweeps probe increasingly fine granularity.

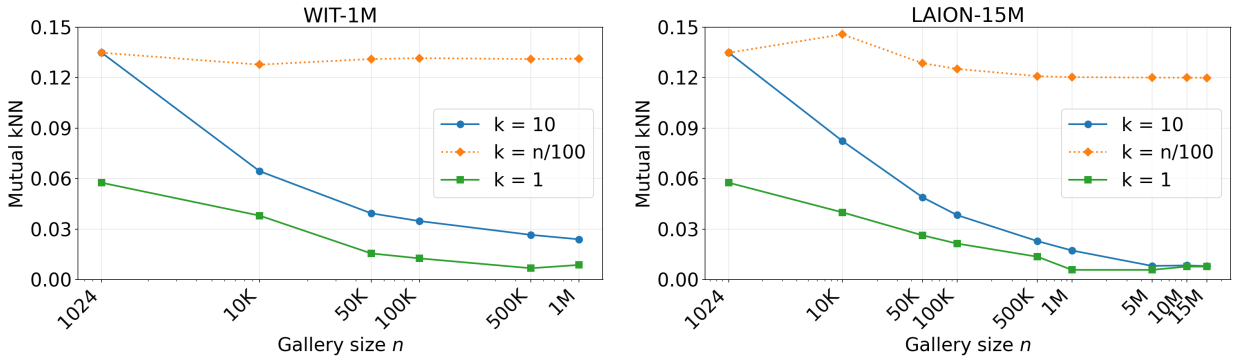


Figure 4: Scaling the gallery size to 1M (WIT) and 15M (LAION) shows a large drop in mutual kNN alignment for k=1 and k=10 for DINOv2 and OpenLlama features. Because $\frac{k}{n}$ shrinks as n grows, we probe progressively finer granularity. Mutual kNN alignment for $k = \frac{n}{100}$ is stable.

Crucially, this decay is not a metric artifact. Under identical 1M and 15M densification, same-modality baselines (e.g. OpenLlama-3B vs. 13B, or DINOv2 vs. CLIP visual) keep strong alignment (Fig. S1, $R_{k\text{NN}} \in [0.35, 0.62]$), whereas cross-modal pairs collapse to near zero. This is consistent with cross-modal misalignment reflecting genuine organizational divergence. At increasing data density, candidates get semantically closer, placing greater demand on models to agree on fine-grained distinctions.

Conversely, alignment at $k = \frac{n}{100}$ remains stable across scales. This establishes broad category-level correspondence, expected from overlapping internet data, but convergence to the *same* representation additionally requires agreement on fine-grained structure, which is exactly what decays. While Gröger et al. (2026) reported local neighborhood persistence, their effective neighborhood size ($k \approx n/100$) falls entirely within this coarse regime. When probing genuinely fine-grained neighborhoods ($k \ll n/100$), local agreement collapses.

We next re-examine the trend reported by Huh et al. (2024), where stronger LLMs exhibit higher alignment with visual features (DINOv2). On WIT-1024 ($k = 10$), this trend is clear (Fig. 2b). However, at the WIT-1M scale, the gap between LLMs narrows and the trend disappears for fixed small k. The trend reappears on WIT-1M when restoring the proportional coarse ratio $k = \frac{n}{100}$ (Fig. S2). As $k = 10$ on $n = 1024$ represents a coarse metric ($k \approx n/100$), the original evidence in Huh et al. (2024) tracks coarse overlap rather than fine-grained representational convergence.

The nearest-neighbor examples in Fig. 5 further illustrate this. Matches at small gallery sizes often disappear as candidates are added, with each modality finding better but divergent neighbors. The matches at 1M and 15M scale are mostly near-duplicates that our deduplication missed (e.g. a crop shifted by a few pixels). We show additional visualizations in Figs. S18 to S21.

Observation 1: Mutual kNN alignment at fixed small k decays for denser galleries for cross-modal pairs, while same-modality pairs remain significantly more stable under identical densification. The reported trend that stronger language models align better with vision is a property of the coarse regime. It persists at $k = \frac{n}{100}$ but vanishes at fixed $k = 10$ on WIT-1M.

3.2.2 Text-audio and text-video alignment

To check that our findings are not specific to text-image, we repeat the scaling analysis for a video encoder (VideoMAE-v2 (Wang et al., 2023a)) on the PVD-100k dataset, a 100k subset of PVD (Bolya et al., 2025; Wang et al., 2019), and an audio encoder (Dasheng from Dinkel et al. (2024)) on a 100k subset of LAION-Audio-630k (Wu et al., 2023).

Fig. 6 shows that the same patterns observed for text-image alignment hold across these modality pairs. Here, the LLM used is Gemma-2-9b (Gemma Team, 2024). Alignment at fixed small k drops with gallery size, while alignment at k=$n/100$ remains stable across scales. The trend that stronger LLMs align better with the non-text modality (Fig. 6c) is weak. We observe a slight positive slope for text-video and an essentially flat evolution for text-audio. This measurement sits in the coarse regime, where the text-image capability

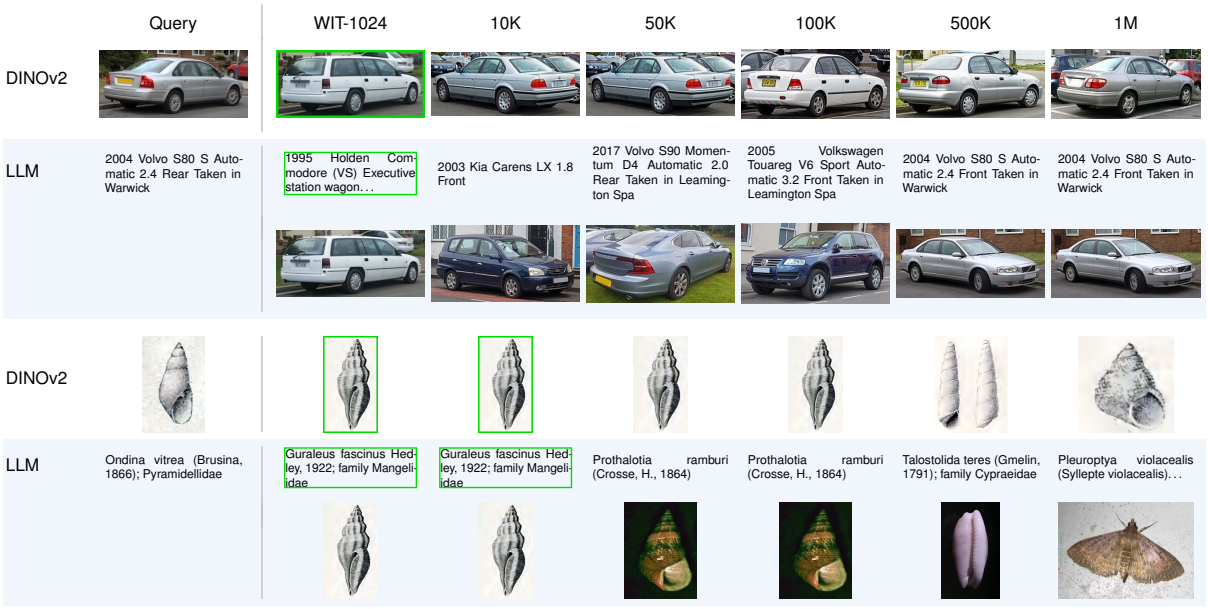
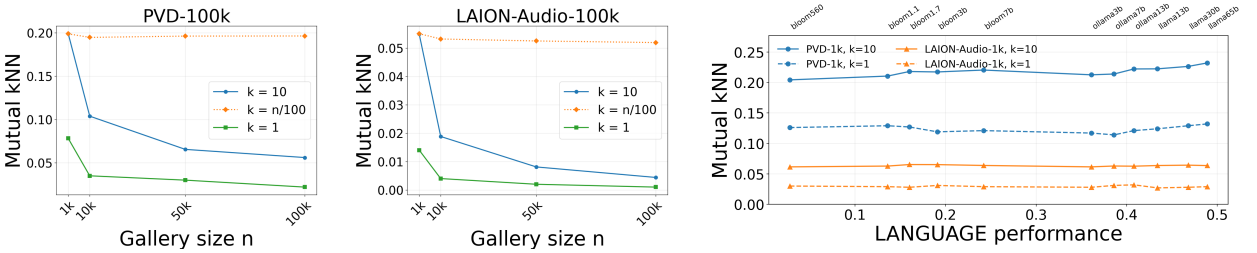


Figure 5: Nearest-neighbor (k=1) examples with DINOv2 and OpenLlama across gallery scales on WIT-1M. Captions are shown with corresponding images. Mutual kNN matches across modalities are framed in green. *At larger scales, each model finds closer but most often different matches.*



(a) Text-video alignment. (b) Text-audio alignment. (c) Alignment vs. LLM performance.

Figure 6: Cross-modal alignment beyond text-image. (a, b) Mutual kNN text-video (PVD-100k (Bolya et al., 2025; Wang et al., 2019)) and text-audio alignment (LAION-Audio-100k (Wu et al., 2023)) scales similarly to the text-image setting. (c) Alignment vs. LLM performance at 1k: weak trend for video, flat for audio.

trend does persist. The flat audio trend may reflect the terseness of the audio captions, consistent with our text–image finding that terse text caps cross-modal alignment regardless of which language model reads it (Fig. 10). We note the narrower scope of this experiment relative to our text-image analysis in terms of models and gallery sizes.

3.2.3 Trend check: Are the predictions from Huh et al. (2024) holding up so far?

Huh et al. (2024) predict that as LLMs become stronger, their representations align more with vision representations. This claim is evaluated using three proxies for language performance: HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and (1 − bitsperbyte) (similar to Huh et al. (2024)).

In this section, we revisit this trend analysis for an extended set of models and benchmarks. We follow the original experimental setup from Huh et al. (2024) to evaluate 55 LLMs, spanning from BLOOMZ (Muennighoff et al., 2023) to recently released models. We use the ARC Challenge (Clark et al., 2018), MMLU (Hendrycks et al., 2021), and LogiQA2 (Liu et al., 2023) benchmarks. These probe scientific question answering (ARC), general knowledge (MMLU), and logical reasoning (LogiQA2), complementing GSM8K's arithmetic reasoning. We present results for models that surpass Llama-3-70B (Grattafiori et al., 2024) (the strongest model in Huh

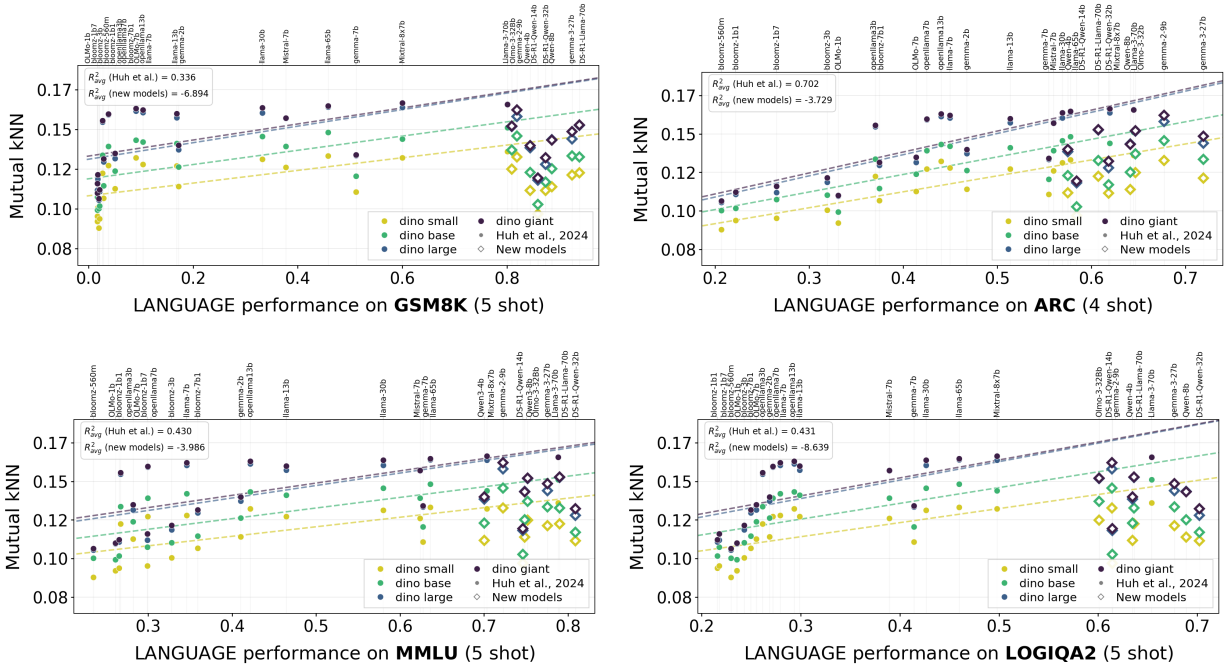


Figure 7: Testing whether the alignment-LLM performance trend from Huh et al. (2024) (dashed lines), tested on WIT-1024, holds for recent LLMs on ARC, GSM8K, MMLU, and LogiQA2. Recent LLMs (diamonds) do not follow the trend extrapolated from the models in Huh et al. (2024), alignment with DINOv2 saturates.

et al. (2024)) on at least one benchmark, testing whether the alignment-capability trend continues. The complete results for all models and benchmarks are included in Section F.

Our observations depend on the benchmark used to measure language-model performance. For the benchmarks used in Huh et al. (2024), such as HellaSwag and Wikitext, the trend partially extends to newer models, with positive R^2_{new} of 0.30 and 0.49. This partial continuation is specific to the local metric. Under calibrated RSA and CKA, HellaSwag, and Wikitext also fail to extrapolate (Section F.4). Those benchmarks measure next-token prediction and commonsense language understanding, closely tied to the autoregressive pretraining objective. However, these benchmarks are increasingly saturated for recent LLMs. We therefore focus on broader reasoning and knowledge benchmarks.

On reasoning benchmarks (Fig. 7), we observe that recent models do not continue the scaling lines extrapolated from the model set from Huh et al. (2024). Instead, their points do not follow the predicted alignment trend and hint at saturation with respect to DINOv2 features. The R^2 ranges from -1.75 to -0.58. The linear fit from the base models is a worse predictor of alignment for recent models than simply predicting their mean, i.e., the trend does not extrapolate. Restricting the new-model set to base pretrained checkpoints (excluding instruction-tuned and reasoning-distilled models) leaves the saturation unchanged (R^2_{newbase} ranges from -5.11 to -1.34, see Table 4), so the failure to extrapolate is not an artifact of the post-training regime.

We note that the trend check holds the vision side fixed. A fixed encoder can itself cap alignment (Section 4), but the saturation appears against all four DINOv2 variants up to ViT-G/14 (Figs. S13 and S14). The joint scaling of both modalities is left to future work.

Observation 2: The alignment–capability trend saturates on reasoning benchmarks. Recent LLMs do not continue the scaling lines extrapolated from Huh et al. (2024) for base and post-trained models alike.

3.2.4 What happens when the data is not bijective?

In practice, the relationship between modalities, such as image and text, is inherently many-to-many: a single image can be described by countless text descriptions, and a single text caption can correspond to a large set

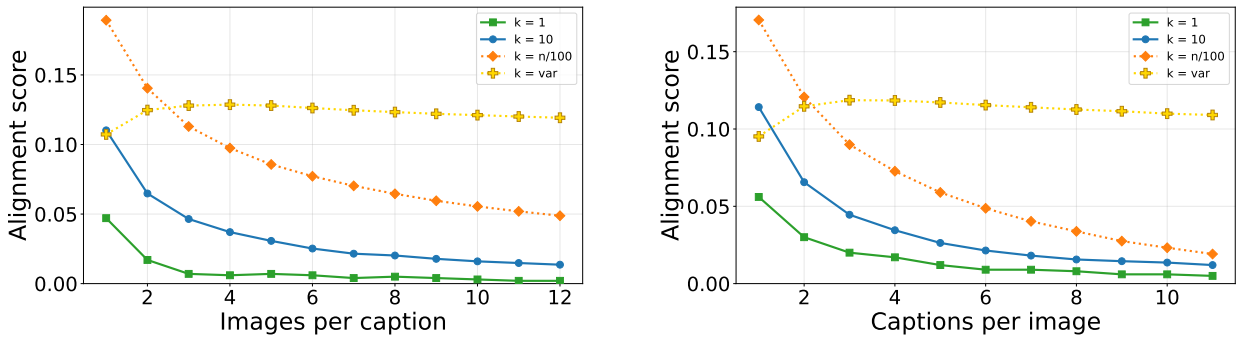


Figure 9: Effect of relaxing the bijective assumption on text-image alignment, using the CycleReward dataset (Bahng et al., 2025). We densify one modality by adding more images per caption (left) or captions per image (right) while keeping the other fixed. Mutual kNN alignment decreases for both k=1, k=10, and $k = \frac{n}{100}$. Our variable k adaptation of mutual kNN remains stable.

of visually distinct images. More fundamentally, modalities often differ in information content. Specifically, images encode spatial and textural structure that text captures only to a limited extent. On the other hand, text encodes abstraction, negation, and compositional semantics that images do not. One could, in principle, bridge this gap trivially. For instance, one could encode pixel values as text or render captions as images and establish a bijection between those. These preserve the information, but the inductive structure (the specific properties that make each modality useful) of each modality is lost.

To test what happens when bijectivity is relaxed, we use the CycleReward dataset (Bahng et al., 2025) which pairs each real sample with multiple synthetic candidates. The I2T subset contains 11 generated captions per real image, and T2I consists of 12 synthetic images for each text prompt. This directly breaks the bijection, i.e. one-to-one matching, that our earlier analysis assumes. We evaluate mutual kNN by densifying one modality at a time: for T2I we keep the text fixed and increase the number of generated images per prompt, and for I2T we keep the image fixed and add generated captions. For illustration, let us consider the T2I experiments where the closest neighbor in the densified modality is more likely to be a similar image that is associated with the same caption, while in the sparse text space the NN will be a caption for a different image (see Fig. 8).

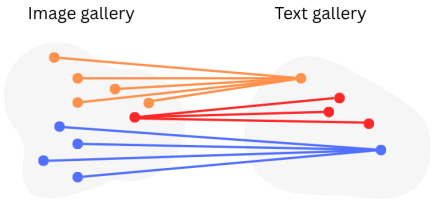


Figure 8: Non-bijective correspondence. The nearest neighbor of a text caption for one image (blue) is a caption for a different image (red). The nearest image neighbor for a given image may be another image with the same caption.

We adapt the mutual kNN metric so that a match is counted when the retrieved item corresponds to the source sample, even if it is not the exact same caption or image. In Fig. 9, we see that the mutual kNN scores still decrease as the one-to-one assumption is relaxed. The convergence evidence in Huh et al. (2024) rests on a bijective pairing that real-world multi-modal data rarely satisfies, and the mutual kNN metric does not distinguish between actual misalignment and differences in density across modalities. Hence, we adapt the mutual kNN metric to a variable k that measures a constant neighborhood radius around a query as the gallery grows, whilst collapsing non-bijective data into bijections when computing the metric (similar to the fixed k extension, see Section C.1).

Observation 3: As the one-to-one assumption is relaxed, the standard mutual kNN variants decay. On non-bijective data, it reflects correspondence density as much as alignment, revealing a limitation of the evaluation protocol.

4 What kind of structure do vision and language models share?

Section 3.2 established that the agreement surviving densification is coarse. Here, we characterize both what is shared and what is not, on the ImageNet validation set (Deng et al., 2009), which operates in a retrieval

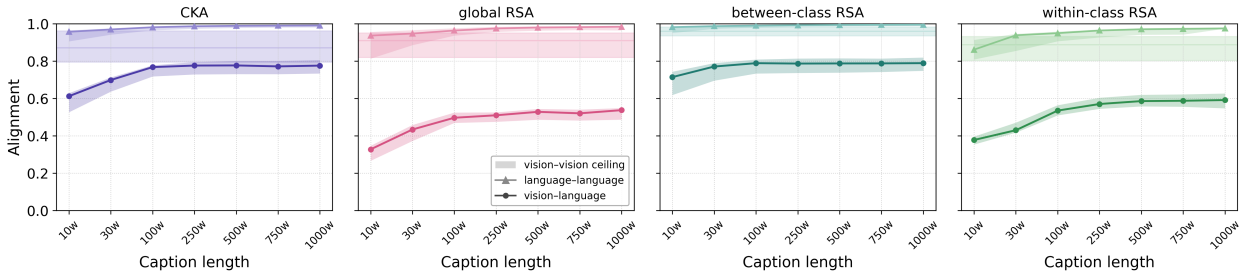


Figure 10: Unimodal alignment ceilings for different caption lengths on the ImageNet validation set, for CKA, global RSA, and between- and within-class RSA. Vision–vision (horizontal band) and language–language (triangles) ceilings stay high across caption length, while the vision–language (circles; DINOv2-base against 12 LLMs) band saturates by 100–250 words well below (particularly for RSA).

regime comparably dense to WIT-1M (see Table 1). The ImageNet class labels additionally let us isolate fine within-class ordering.

Using global instead of local metrics. We complement fixed-k mutual kNN with four additional metrics. RSA is the Spearman correlation between the two models' pairwise cosine-distance vectors over the same samples. Specifically, within-class RSA computes it separately inside each ImageNet class and measures the fine same-category ordering. Linear CKA compares the dominant directions of variation (Kornblith et al., 2019). The variable-k mutual kNN sets each query's neighborhood size from its own local density (all gallery items within the query's self-NN distance) and scores neighborhood IoU. The differences between these metrics provide information about agreement at different granularities. Details are in Section C.

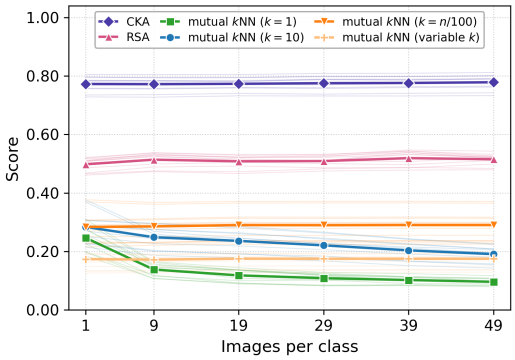


Figure 11: Cross-modal alignment on the ImageNet validation set (500-word captions). As the gallery grows, fixed-k mutual kNN decays, while k=n/100, variable-k, RSA, and CKA are stable.

Fig. 11 considers the galleries from 1 to 49 images per class from the ImageNet validation set with 1000 queries (one per class), and 500-word per-image Gemini captions (Section G.2). Mutual kNN at k=n/100 and variable-k, RSA variants, and linear CKA stay stable. Consequently, they read the geometry the models impose rather than being gallery-dependent measures. In the following, we focus on RSA and CKA.

The gap between cross-modal and unimodal alignment is organizational. We compare the calibrated cross-modal scores to how strongly two good models agree when they observe the *same* modality. The unimodal ceiling is a band (Fig. 10): vision pairs reach RSA ≈ 0.95 same-family and ≈ 0.82 cross-family (DINOv2 $\times$ DINOv3 (Siméoni et al., 2025)), and 6 LLMs share the within-class fine ordering at 0.94–0.98 across families. The fine structure is successfully encoded in each modality. However, cross-modal agreement behaves differently. If missing descriptions were the bottleneck, richer captions would close the gap. Instead, the cross-modal band stalls by 100–250 words, while unimodal agreement is high at every caption length (Fig. 10). The size of the cross-modal gap depends on the granularity a metric reads. Cross-modal alignment reaches 83–92% of the vision ceiling in CKA, which is dominated by the coarse, shared directions of variation, but only 51–58% in RSA, which grades the full ordering.

The modalities agree on the coarse layout and then organize the fine structure differently. Truncating both representations to their top j eigenmodes locates this directly (Fig. 12). Cross-modal RSA peaks at j=3, where three modes carrying 80% of the CKA weight already give the peak agreement (0.75), and erodes to 0.55 as the tail is added. The tail is not noise. Each model's agreement with its own full representation keeps rising over the same range (Section B.3), so the tail carries structure each modality orders consistently and the other never matches.

The gap is largest where the structure is most fine-grained, and is smaller when measuring coarse structure. Within-class cross-modal RSA reaches 77–85% of the vision ceiling, against 51–58% for the global ordering, saturating by ∼100 words (Fig. 10). Collapsing each class to its centroid leaves only between-class geometry. Between-class coarse cross-modal RSA is 0.79 against the weakest vision pair's 0.94. Coarse alignment has the smallest gap, but not a closed one.

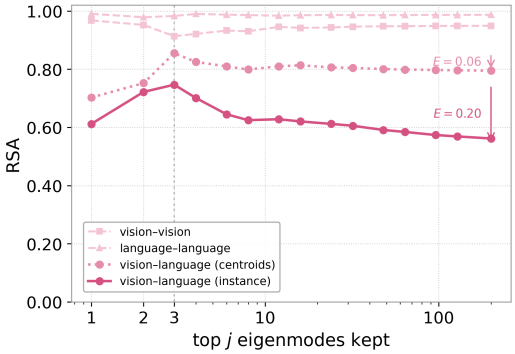


Figure 12: Truncated RSA (top-j eigenmodes) on ImageNet (validation). Cross-modal RSA erodes with more modes, unimodal is flat. Collapsing to class centroids shrinks erosion (E). Agreement is in the top 3 eigenmodes, disagreement is within classes.

Using von Uexküll's *Umwelt*, we distinguish informational ceilings from organizational alignment. Short captions resemble an informational ceiling by omitting visual details. Enriching them narrows the cross-modal gap while the unimodal ceilings stay flat (Fig. 10). However, the narrowing saturates by 100–250 words, and even with multi-paragraph descriptive captions, cross-modal RSA stalls well below unimodal levels. Vision and language models occupy distinct *Umwelten*.

Observation 4: The cross-modal gap is organizational. Both modalities represent fine structure consistently but arrange it differently, so the gap is smallest for CKA, which reads coarse shared structure, and largest for RSA, which grades the full ordering.

5 Discussion and future work

The Platonic Representation Hypothesis suggests that models trained on different modalities align and eventually converge toward the same representation of reality as they scale. Our results separate claims about the representations themselves from limitations of the original experimental evidence.

The drop of fixed-k mutual kNN alignment as data scales traces how far cross-modally shared structure extends. Same-modality pairs remain far more stable under identical densification. Even in the original small-scale regime, the reported trend that stronger language models align better with vision does not extrapolate: for recent models, alignment saturates on reasoning benchmarks. The reduction of mutual kNN under non-bijective correspondences, by contrast, is a limitation of the protocol: it conflates genuine misalignment with the many-to-many structure of real image-text data. Measuring what the modalities share therefore requires separating coarse from fine structure.

On ImageNet, class labels provide this separation. Cross-modal agreement approaches the same-modality ceiling at coarse level, but falls short on the fine ordering within a class. Huh et al. (2024) anticipate that non-bijective observation functions cap alignment, but leave open whether the cap matters. Our measurements show that it does: richer captions increase alignment, but the gain saturates well before the longest captions we generate, leaving cross-modal agreement below the same-modality ceiling at both coarse and fine granularity.

The text and image *Umwelten* overlap in coarse structure but diverge in fine-grained organization. A cross-modal metric therefore measures translation between two differently organized maps, not progress toward a common form.

Future work: in search of bijection. Prior work on the Platonic Representation Hypothesis evaluates alignment under a one-to-one correspondence assumption between modalities. However, real-world image–text data is inherently many-to-many, and the extent to which any approximate bijection exists at the level of representations remains unclear. A key direction for future work is to directly test whether language can serve as a lossless bottleneck for image reconstruction (i.e. an image-text-image autoencoder). Recent round-trip image–text diffusion (Guo et al., 2026) and panoptic captioning (Lin et al., 2025) provide frameworks for such a test. If a lossless bottleneck proves elusive in realistic settings (e.g. text bottlenecks under a thousand words), the next question is which part of the text-image space (intersection of a Venn diagram) admits a bijection, and how to disentangle it from the rest.

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Supplementary Material

Back into Plato's Cave: Examining Cross-modal Representational Convergence at Scale

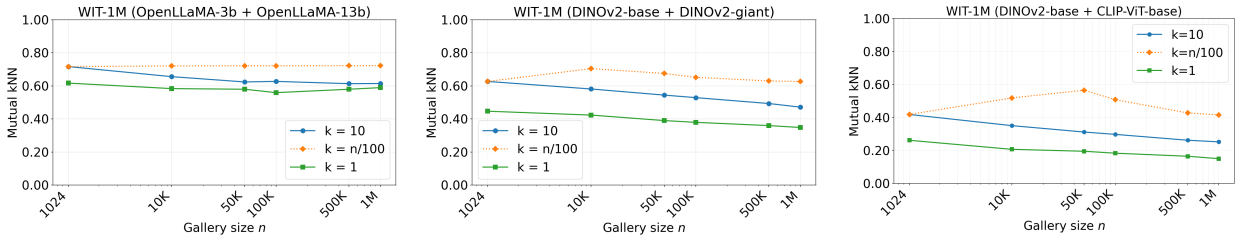
A Is the drop in mutual kNN alignment at scale caused by the metric, layer selection, caption quality, or by specific model choices?

In this section, we provide additional experiments to verify that the main findings in the paper are not due to a confounding variable.

A.1 Sanity check: does mutual kNN inherently drop at scale even within modalities?

We test whether the alignment drop with increasing gallery size (Fig. 4 in the main paper) is merely an artifact of the mutual kNN metric being harder at scale.

Specifically, we measure within-modality alignment for two pairs of models: two language models of different scales (OpenLlama-3b and OpenLlama-13b), and, separately, two vision model pairs (DINOv2-base / DINOv2-giant and DINOv2-base / CLIP-B16). If mutual kNN alignment collapses for dense galleries regardless of the models being compared, the cross-modal drop observed in the paper would be uninformative. If within-modality alignment remains more stable, the cross-modal drop is meaningful.



(a) OpenLlama-3b and OpenLlama-13b

(b) DINOv2-base and DINOv2-giant

(c) DINOv2-base and CLIP-B/16

Figure S1: Unimodal mutual kNN alignment as a function of gallery size on WIT-1M, for same-family pairs (a, b) and a cross-family vision pair with no shared training recipe (c). *In contrast to cross-modal alignment (Fig. 4), unimodal alignment, including for the cross-family pair, remains significantly more stable across scales.*

As shown in Fig. S1, unimodal alignment remains much more stable across gallery sizes than the cross-modal alignment reported in the main paper. For the OpenLlama pair, mutual kNN at k=1 stays between $[0.59, 0.62]$, and for the DINOv2 pair between $[0.35, 0.45]$, across all gallery scales. Cross-family alignment is lower than same-family, as expected, but remains more stable than cross-modal alignment. **Unimodal mutual kNN decays by at most $2\times$ over the same range where cross-modal decays by $50\times$ (k=1). The cross-modal decay reflects that the structure shared across modalities does not extend to the fine granularity probed by small k at large n.**

We additionally verify that the alignment trend occurs at the coarse $k{=}\frac{n}{100}$ level: Fig. S2 compares the WIT-1024 trend at k=10 with the WIT-1M trend at $k{=}\frac{n}{100}$, and the two largely match. This is consistent with the original evidence using mutual kNN reflecting coarse semantic-category overlap rather than fine-grained representational convergence.

A.2 Sanity check: is the alignment drop an artifact of the layer pair selected at WIT-1024?

For our experiments, we select the best layer pair on the 1024-sample subset of WIT and reuse this pair across larger-scale evaluations. Since our central claim concerns how alignment behaves as the gallery densifies, we test whether the observed patterns persist when the layer pair is re-selected at each gallery scale.

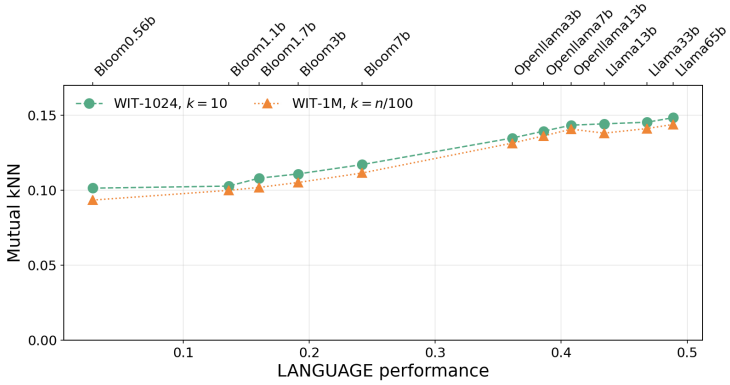
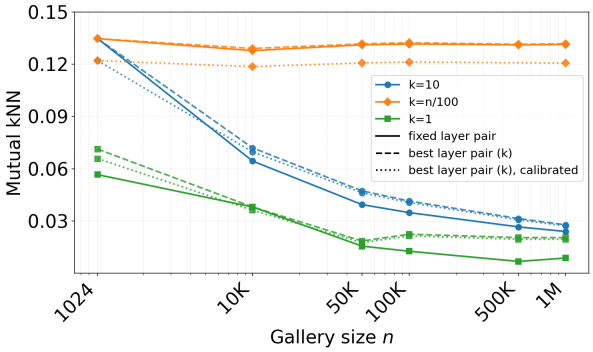


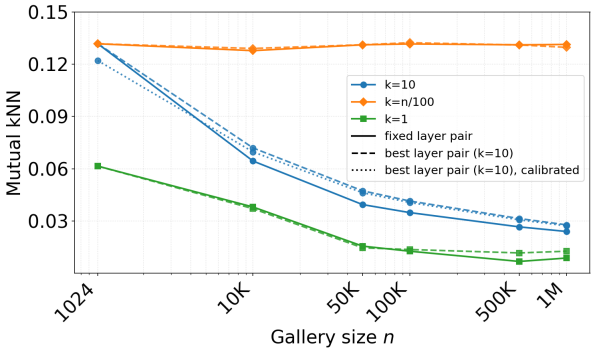
Figure S2: Alignment vs. LLM performance: the WIT-1024 trend at k=10 compared to WIT-1M at k=$\frac{n}{100}$. *Unlike the fixed-k case (Fig. 2b in the main paper), the capability trend persists for k=n/100, consistent with stable coarse overlap rather than fine-grained convergence.*

Specifically, we re-run the layer search at each gallery size in $\{1024, 10\text{K}, 50\text{K}, 100\text{K}, 500\text{K}, 1\text{M}\}$ and report the resulting mutual kNN alignment. Fig. S3 shows the comparison: (a) re-selecting layers per scale and per k (i.e., the optimal layer pair for k=1, k=10, and k=$\frac{n}{100}$ separately at each gallery size), and (b) re-selecting layers per scale at fixed k=10 and applying the same layer pair to other k values. Exhaustive layer-pair search inflates alignment scores by selection. We follow the null calibration proposed by Gröger et al. (2026), subtracting the expected alignment under permuted image-text pairings to correct for this bias (shown as dotted lines).

In both cases, the alignment trend with increasing gallery size is preserved. **This indicates that the alignment drop for fixed small k reported in the main paper does not suffer from an artifact of the layer pair being fixed at WIT-1024 scale.**



(a) Per-scale layer re-selection performed independently for each k. The alignment trends are preserved.



(b) Per-scale layer re-selection performed at k=10 and applied to all k values. The same trend is observed.

Figure S3: Cross-modal mutual kNN alignment on WIT-1M with the layer pair re-selected at each gallery scale, rather than fixed at WIT-1024. Dotted lines show calibrated scores following Gröger et al. (2026). *The qualitative scaling trend is unchanged, indicating that the alignment drop is not driven by suboptimal layer choices.*

A.3 WIT-1M-recap: Is the alignment drop caused by poor captions?

One might hypothesize that the alignment drop at scale is driven by the low quality of the WIT captions rather than by a fundamental cross-modal difference. To test this, we recaption WIT-1M using gemini-3-flash-preview as described in Section G.2. The resulting WIT-1M-recap dataset contains visually detailed descriptions

of around 500 words per image. As shown in Fig. S4, mutual kNN alignment still drops with gallery size. More detailed captions give overall higher mutual kNN scores, but do not prevent the decline in scores. **This suggests that caption quality is not the primary driver of the mutual kNN alignment drop**.

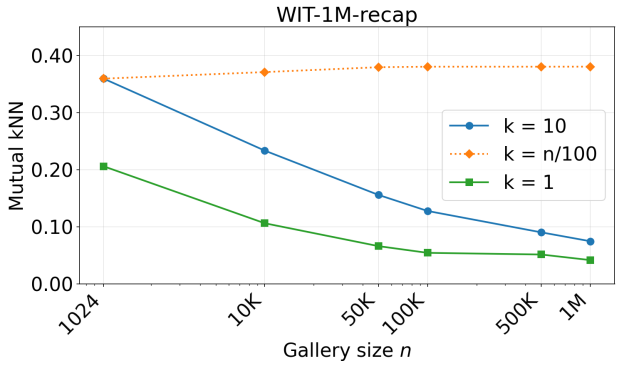


Figure S4: Cross-modal mutual kNN alignment on images recaptioned using gemini-3-flash-preview (WIT-1M-recap) as the gallery grows to 1M samples. *Detailed captions result in overall higher mutual kNN scores, but do not prevent the drop in scores.*

A.4 Mutual cross-modal kNN alignment drops at scale across model pairs

The cross-modal alignment drop reported in Fig. 4 in the paper uses DINOv2-base and OpenLlama-3b. Here, we examine whether similar patterns hold for stronger models. In Fig. S5, we repeat the scaling experiment for two additional model pairs: DINOv2-base with OpenLlama-13b, and DINOv2-giant with OpenLlama-13b.

Replacing DINOv2-base with the stronger DINOv2-giant and OpenLlama-3b with OpenLlama-13b does not change the pattern. We observe that mutual kNN still drops at scale. This is consistent with Fig. 2b in the paper, which shows low alignment scores across different LLMs at WIT-1M scale.

To test whether the pattern is specific to the DINOv2/OpenLlama choice, we evaluate two further axes of variation. First, we vary the vision encoder while keeping OpenLlama-3B fixed (Fig. S6): we replace DINOv2 with Pixio-ViT-B/16 (Yang et al., 2026) and with CLIP-B/16 (image-text contrastive pretraining). Second, we vary the LLM while keeping DINOv2-base fixed (Fig. S7): we pair it with Gemma-7B (Gemma Team, 2024a) and Mistral-7B (Jiang et al., 2023) from different model families. Across all four combinations, the alignment drop at fixed small k persists, while alignment at $k = \frac{n}{100}$ remains stable.

This indicates that the degradation reported in the paper is not specific to that particular choice of models.

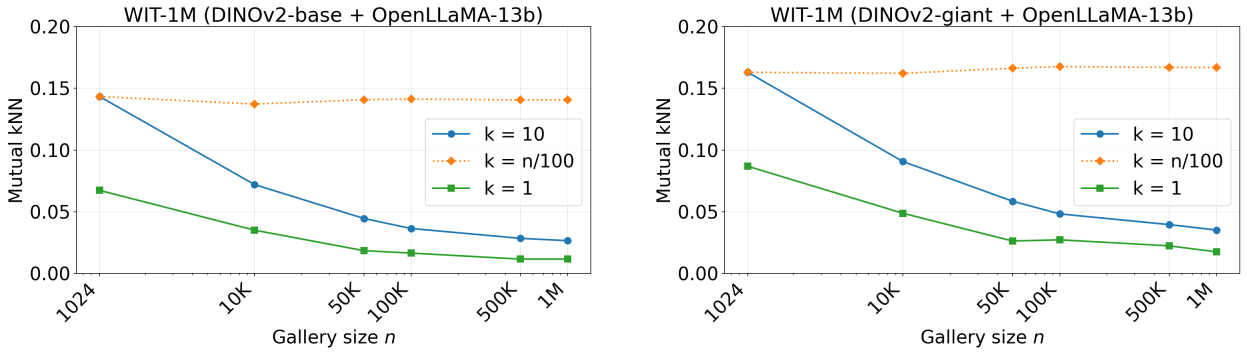
B Additional ImageNet experiments

Here, we verify that the ImageNet validation set used in Section 4 in the main paper serves as a suitable test bed. Furthermore, we confirm that our observations are not limited to our choice of models or metric settings.

B.1 The ImageNet validation set is denser than WIT-1024

A natural question is how the gallery density in our ImageNet experiments compares to the WIT data. As shown in Table 1, nearest-neighbor cosine similarities on the ImageNet validation set are substantially higher than on WIT-1024 and comparable to WIT-1M. Even for only one image per class in the gallery (ipc=1), the ImageNet validation set provides a denser retrieval setting than WIT-1024.

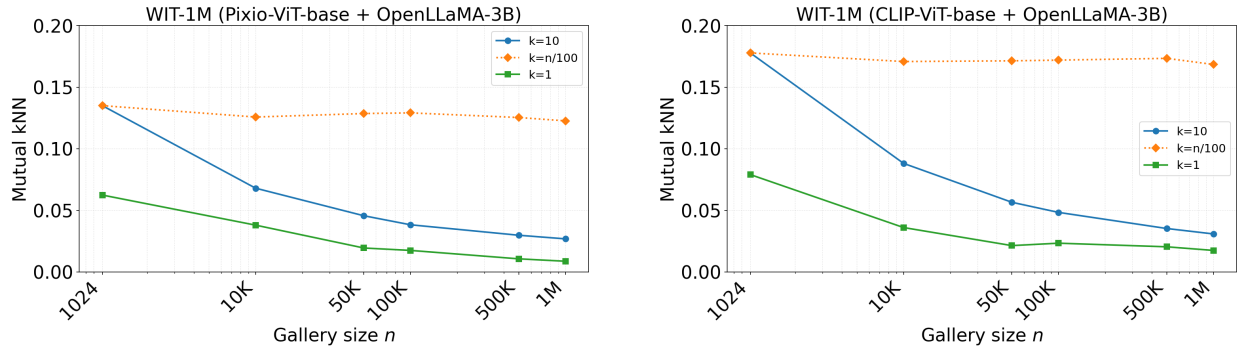
This confirms that the ImageNet experiments operate in a denser retrieval regime comparable to WIT-1M, making this a meaningful test bed.



(a) DINOv2-base and OpenLlama-13b

(b) DINOv2-giant and OpenLlama-13b

Figure S5: Cross-modal mutual kNN alignment as gallery grows from WIT-1024 to WIT-1M for additional, stronger model pairs. Replacing DINOv2-base with the stronger DINOv2-giant and OpenLlama-3b (Fig. 4 in the paper) with OpenLlama-13b does not prevent the drop. *This suggests that the degradation in mutual kNN alignment was not a result of the limitation of any individual model.*



(a) Pixio-ViT-B/16 and OpenLlama-3B

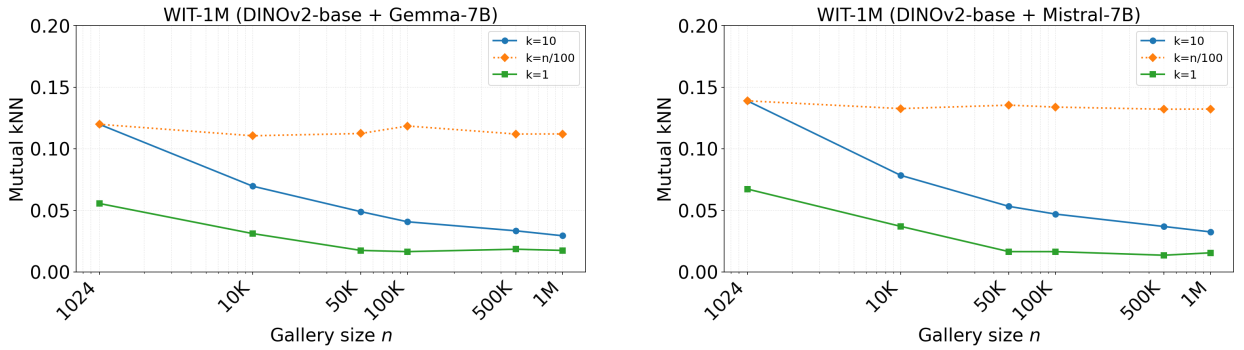
(b) CLIP-B/16 and OpenLlama-3B

Figure S6: Cross-modal mutual kNN alignment as the WIT gallery grows from 1024 to 1M, for vision encoders trained with different objectives (Pixio-style SSL (Yang et al., 2026) and CLIP contrastive image-text pretraining), paired with OpenLlama-3B. *The same scaling pattern holds: alignment at fixed small k drops with gallery size while k=n/100 stays flat.*

B.2 Ceiling bands: label-free replication (WIT-1024) and the mutual kNN metrics

WIT-1024. WIT-1024 has no labels, and the ImageNet picture replicates without them (Fig. S8): the language ceiling is flat and high at every Gemini length (RSA 0.92–0.99) while cross-modal alignment saturates by ∼250 words far below it (RSA 0.48 → 0.64, CKA 0.66 → 0.82). The original WIT-1024 captions sit below every Gemini length and are the one caption type that drags down the language ceiling itself (cross-family pair 0.75): noise in the observation, not in the models. Since the calibrated re-examination of Gröger et al. (2026) evaluates on these original terse captions, the alignment it can measure is capped by the observation before calibration enters. On the same data, cross-modal alignment sits substantially higher once captions are rich.

The local metric. The mutual-kNN bands (Fig. S9 for ImageNet, right panel of Fig. S8 for WIT-1024) repeat the cross-modal pattern of the global metrics, with two properties specific to reading top-k neighborhoods. The vision–vision band is wide: local agreement is graded by model scale and family rather than forming a single ceiling, with same-family pairs at the top and small-model and cross-family pairs at the bottom. And the language–language band lies above the vision band: LLMs reading the same captions share their local neighborhoods more than two differently trained vision backbones share theirs.



(a) DINOv2-base and Gemma-7B

(b) DINOv2-base and Mistral-7B

Figure S7: Cross-modal mutual kNN alignment as the WIT gallery grows from 1024 to 1M, with DINOv2-base paired with LLMs from different families (Gemma and Mistral). *The scaling pattern is consistent across LLM families: drop at fixed small k, stability at $k{=}n/100$.*

Table 1: Nearest-neighbor proximity across gallery sizes. ImageNet, even with only one image per class in the gallery (ipc=1), has neighbor proximities comparable to WIT-1M, confirming that it operates in a similarly dense retrieval regime.

Gallery	Model	Dim	k=1	k=10
WIT-1024	DINOv2-base	768	0.799	0.717
WIT-1024	OpenLlama3b	3200	0.502	0.400
WIT-1M	DINOv2-base	768	0.906	0.888
WIT-1M	OpenLlama3b	3200	0.757	0.701
ImageNet ipc=1	DINOv2-base	768	0.823	0.763
ImageNet ipc=1	DINOv2-giant	1536	0.609	0.496
ImageNet ipc=1	OpenLlama3b	3200	0.928	0.904
ImageNet ipc=1	LLaMA-65B	8192	0.890	0.858
ImageNet ipc=49	DINOv2-base	768	0.887	0.861
ImageNet ipc=49	DINOv2-giant	1536	0.749	0.690
ImageNet ipc=49	OpenLlama3b	3200	0.954	0.944
ImageNet ipc=49	LLaMA-65B	8192	0.926	0.912

Neighborhood composition. The band width is set by what neighborhoods are made of. Gallery size is irrelevant (vision–vision overlap is flat from 1k to 49k gallery images), but replaying the WIT protocol on ImageNet with one image per class reproduces the wide WIT band almost cell for cell (0.48–0.68), while twenty images per class compress it to 0.73–0.82. With class siblings available every model fills the top-10 with siblings and the pairs agree; without them the metric reads fine cross-item distinctions, where vision pairs separate by scale and family. None of this affects the cross-modal comparison, which stays far below both bands at every caption length.

The ceiling-band structure also emerges without using class labels. For WIT-1024 the language ceiling stays at 0.92–0.99 **at every caption length while cross-modal alignment saturates at** 0.64**. The wide vision m-kNN band is neighborhood composition: one-image-per-class ImageNet reproduces it, and class-sibling density restores a tight band.**

B.3 Cross-modal agreement lives in the top three eigenmodes

Section 4 reports the truncated-rank analysis locating cross-modal agreement in the top three eigenmodes (Fig. 12). Here we give the formal setup, the self-resolvability control, the CKA companion, and the population-level statistics.

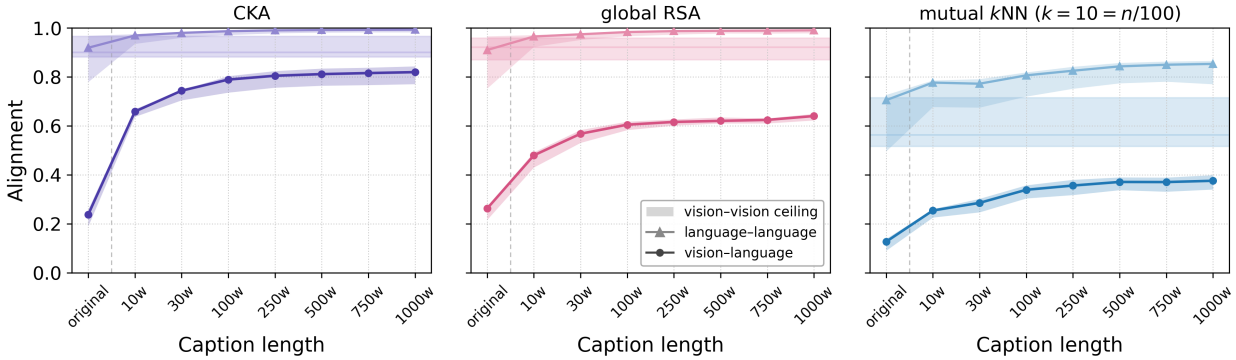


Figure S8: Alignment bands on WIT-1024 across caption types: original captions, then Gemini captions of increasing length ($n = 1024$; all cells calibrated, own 2D layer search; curves are medians over pairs, shading the min–max range). Horizontal band: six vision–vision pairs (caption-independent). Triangles: four language–language pairs per caption type. Circles: twelve validation LLMs against DINOv2-B. The mutual kNN panel is reference-only at this sample size.

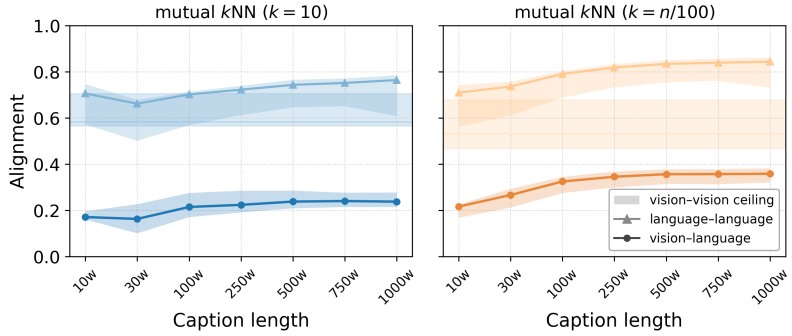


Figure S9: Mutual-kNN ceiling bands across caption lengths on the ImageNet validation set, the local-metric companion of Fig. 10 (same cells and calibration; curves are medians over pairs, shading the min–max range).

CKA yields much higher cross-modal scores than RSA. We can measure the source of this difference explicitly.

Let $X \in \mathbb{R}^{n \times D_X}$ and $Y \in \mathbb{R}^{n \times D_Y}$ be the column-centered feature matrices of two encoders on the same n samples, with PC variances λ_i and score directions $u_i \in \mathbb{R}^n$. The score directions live in the shared sample space, so mode cosines $u_i^{X\top} u_j^Y$ are defined across modalities even though $D_X \neq D_Y$. Linear CKA weights the agreement of a mode pair by its variance product, $\mathrm{CKA} = \sum_{ij} \hat{\lambda}_i^X \hat{\lambda}_j^Y (u_i^{X\top} u_j^Y)^2$ with $\hat{\lambda} = \lambda / \|\lambda\|_2$, so its mass concentrates on the top few modes. Rank-based RSA instead counts every pair-order decision equally, and near-tied orderings are decided by the low-variance tail. To locate where cross-modal agreement lives, we truncate both sides to their top j modes and track $\rho_\times(j)$, the Spearman correlation between the two models' pairwise cosine-distance vectors computed from the truncations X_j and Y_j, with $\rho_{\mathrm{self}}(j)$ the same statistic between X_j and the full X. How much of a model's own final ordering do its top j modes already determine? We report the peak $j^* = \arg\max_j \rho_\times(j)$ and the erosion $E = \rho_\times(j^*) - \rho_\times(\infty)$ (Fig. S10).

At the RSA-optimal instance layers (500-word captions, balanced $n = 5000$), the cross-modal pair (DINOv2-B $\times$ OpenLLaMA-3B) peaks at $j^* = 3$. The top three eigenmodes carry 27%/31% of the two sides' variance and 80% of the CKA weight. Adding the remaining modes erodes the agreement, down to 0.55 at full rank ($E = 0.20$, Fig. S10 left). The tail modes are not noise: self-resolvability is only 0.71/0.69 at j^* and keeps rising, so the modes added beyond the threshold carry real structure that each model orders consistently and the other modality never matches.

On the identical truncations CKA hardly changes (Fig. S10 right). Truncated CKA peaks at the same $j^* = 3$ and erodes by 0.01 (0.00 on centroids), the measured form of the weighting asymmetry.

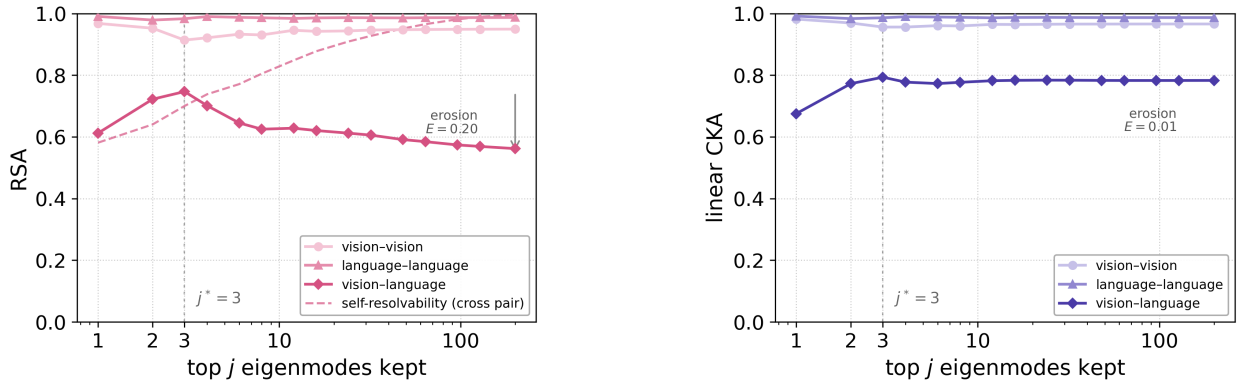


Figure S10: Truncated-rank RSA (left) and truncated-rank CKA (right) between the top-j eigenmode reconstructions of both encoders (instance level, 500-word captions, balanced $n = 5000$; RSA-optimal layers). The cross-modal curve peaks at $j^* = 3$ and erodes under RSA ($E = 0.20$) but not under CKA ($E = 0.01$); unimodal pairs erode under neither. Dashed: self-resolvability of the cross pair, the Spearman agreement between the pairwise distances of each side's truncation and those of its own full representation (averaged over the two sides), which keeps rising where the cross-modal curve falls.

The unimodal controls show negligible erosion at the same layers and sample with $E = 0.02$ for vision $\times$ vision and $E = 0.003$ for language $\times$ language, their curves within 0.05 of their full-rank value at every j and above the cross-modal curve even at its peak. The peak-then-erode shape is a property of the cross-modal pair, not of the truncation. The pattern observed in Fig. 12 is population-level, not a property of one pair: across all twelve validation LLMs, $j^* = 3$ in 23 of 24 cells at both granularities, with instance-level erosion 0.10–0.25 and centroid-level erosion 0.05–0.08 throughout.

Cross-modal rank agreement peaks at $j^* = 3$ eigenmodes and erodes by 0.20 as the remaining modes are added. On the same truncations CKA erodes by 0.01, and unimodal pairs by at most 0.02 in either metric. CKA reads the peak, RSA captures the erosion that lives in the fine-grained structure.

C Alignment metrics and calibration

The mutual kNN metric of Section 3 treats every sample as both a query and a gallery item: the neighborhoods $\mathcal{N}_k^{\mathbf{a}}(i)$ and $\mathcal{N}_k^{\mathbf{b}}(i)$ are retrieved from the same n points that are scored. When we densify a gallery, either by adding samples (Sections 3.2.1 and 3.2.2) or by adding valid correspondences per sample (Section 3.2.4), this couples two effects: the neighborhoods of a fixed query change, and the set of queries being averaged over changes as well. To separate the two, we use a *query-to-gallery* (Q2G) variant of the metric that keeps the query set fixed while the gallery grows. All results in Figs. 6, 9 and 11 are computed with this protocol.

Setup. Unless a metric is rank-based, features are prepared following Huh et al. (2024): per-dimension outlier clamping at the q=0.95 quantile, then L2 normalization. CKA and mutual kNN require this step. RSA operates on ranks and is unaffected by it.

Query and Gallery. Let $\mathcal{Q}$ be a fixed query set of q samples and $\mathcal{G} \supseteq \mathcal{Q}$ a gallery of n samples, with features $\mathbf{a}_i \in \mathbb{R}^{d_1}$ and $\mathbf{b}_i \in \mathbb{R}^{d_2}$ from the two models for every $i \in \mathcal{G}$. Galleries are nested, $\mathcal{G}_1 \subset \mathcal{G}_2 \subset \cdots$, and each contains the queries. Feature preparation (0.95-quantile clamping followed by L2 normalization, Section C) is performed once over the full pool rather than per gallery, so the geometry of the two spaces is fixed and only the number of gallery candidates changes between levels. In the scaling experiments of Section 3.2.2, $\mathcal{Q}$ is a fixed random subset of 1k samples and $\mathcal{G}$ grows from 1k to 100k. In the bijection experiments of Section 3.2.4, $\mathcal{Q}$ holds one correspondence per source sample (one caption per image for I2T, one image per caption for T2I) and $\mathcal{G}$ adds the remaining correspondences one at a time.

Mutual kNN (Fixed k). For every query $i \in \mathcal{Q}$, each model retrieves its k nearest neighbors from the gallery, excluding the query itself:

$$\mathcal{N}_k^{\mathbf{a}}(i) = \text{argtopk}_{j \in \mathcal{G} \setminus \{i\}} \, \mathbf{a}_i^\top \mathbf{a}_j, \qquad \mathcal{N}_k^{\mathbf{b}}(i) = \text{argtopk}_{j \in \mathcal{G} \setminus \{i\}} \, \mathbf{b}_i^\top \mathbf{b}_j,$$

and the score is the mean per-query overlap,

$$s_k = \frac{1}{q} \sum_{i \in \mathcal{Q}} \frac{|\mathcal{N}_k^{\mathbf{a}}(i) \cap \mathcal{N}_k^{\mathbf{b}}(i)|}{k}.$$

For $\mathcal{Q} = \mathcal{G}$ this is exactly the mutual kNN score of Section 3. As in the main paper, we report k=1, k=10, and the gallery-proportional k=$\frac{n}{100}$, where n is the size of the current gallery, while query size stays fixed 1000 or 1024 samples (Section 3.2.1). Because the queries are fixed, any change of s_k across gallery levels is due to the neighborhoods of the same queries being filled by closer candidates, not to a change in which samples are being averaged.

Mutual kNN (Variable k). Fixed-k scores probe a progressively finer granularity as n grows, since $\frac{k}{n}$ shrinks. As a density-matched control, we let each query define its own neighborhood radius. For a reference size k_{ref}, the threshold of query i in each space is its similarity to its k_{ref}-th nearest *other query*,

$$\beta_i^{\mathbf{a}} = \text{topk}^{(k_{\text{ref}})}_{j \in \mathcal{Q} \setminus \{i\}} \, \mathbf{a}_i^\top \mathbf{a}_j, \qquad \beta_i^{\mathbf{b}} = \text{topk}^{(k_{\text{ref}})}_{j \in \mathcal{Q} \setminus \{i\}} \, \mathbf{b}_i^\top \mathbf{b}_j,$$

where $\text{topk}^{(k)}$ denotes the k-th largest value. The neighborhoods are all gallery items within that radius,

$$\mathcal{A}_i = \{\, j \in \mathcal{G} \setminus \{i\} : \mathbf{a}_i^\top \mathbf{a}_j \geq \beta_i^{\mathbf{a}} \,\}, \qquad \mathcal{B}_i = \{\, j \in \mathcal{G} \setminus \{i\} : \mathbf{b}_i^\top \mathbf{b}_j \geq \beta_i^{\mathbf{b}} \,\},$$

and, since $|\mathcal{A}_i|$ and $|\mathcal{B}_i|$ are no longer equal, the score is the mean intersection-over-union,

$$s_{\text{var}} = \frac{1}{q} \sum_{i \in \mathcal{Q}} \frac{|\mathcal{A}_i \cap \mathcal{B}_i|}{|\mathcal{A}_i \cup \mathcal{B}_i|}.$$

The thresholds depend only on $\mathcal{Q}$, so they stay fixed while the gallery grows. Adding gallery samples then enlarges both neighborhoods at a rate set by the local density of each space, rather than shrinking the neighborhood fraction as fixed k does. On the smallest gallery ($\mathcal{G} = \mathcal{Q}$) each neighborhood contains exactly k_{ref} items and s_{var} is a monotone rescaling of $s_{k_{\text{ref}}}$; the two statistics only diverge once the gallery is denser than the query set. We use $k_{\text{ref}} \in \{1, 10, \frac{q}{100}\}$, matching the fixed-$k$ settings.

RSA. Global RSA is the Spearman correlation between the two models' condensed pairwise cosine-distance vectors over the same samples (up to 2,000 samples in the images-per-class sweep. The caption-ladder cells use a canonical class-balanced pool, 5 images per class for the layer search and 15 per class for the reported score). Within-class RSA computes the same correlation separately inside each ImageNet class and averages over classes. The layer search uses a canonical 150-class, 30-images-per-class sample and the reported score covers all 1,000 classes at the selected layers.

CKA. The images-per-class sweep uses linear CKA on prepared features (Kornblith et al., 2019). We use RBF CKA on centered, normalized Gram matrices with the median heuristic bandwidth computed per modality, on up to 4,000 samples.

Layer search and calibration. Every model pair receives its own 2D layer search (all vision layers $\times$ all LLM layers). Each reported score s is calibrated against a pairing-shuffle null: captions are randomly re-assigned to images, the statistic is recomputed, and the null is made search-aware by re-running the full layer-grid maximum per permutation. With τ the null's upper α=0.05 quantile over $K = 200$ permutations, we report $s_{\text{cal}} = (s - \tau)/(1 - \tau)$ (Gröger et al., 2026), so chance-level structure reads as zero.

C.1 Duplicated feature rows in many-to-many data

The many-to-many datasets are stored as one row per correspondence. If a caption is paired with L images, its text feature occurs in L bit-identical rows while the corresponding image features differ. Conversely, when an image is paired with L captions, its image feature is repeated while the text features differ (Fig. 8).

A naive row-level implementation of mutual kNN is unsuitable in this setting. In the repeated modality, several copies of the same feature can occupy multiple positions in a neighborhood, e.g. feature's closest neighbors are bit-identical copies. The other modality distinguishes these rows and may instead retrieve several distinct correspondences.

For each modality $\mathbf{m} \in \{\mathbf{a}, \mathbf{b}\}$, let $\mathcal{R}^{\mathbf{m}} \subseteq \mathcal{G}$ contain one representative, the first occurrence of every bit-identical feature group, and let $r^{\mathbf{m}}(i)$ be the representative of the group containing row i. We also let $\pi(j)$ denote the source group of correspondence row j: the shared caption in the T2I setting or the shared image in the I2T setting. For bijective data, every row forms its own source group and $\pi(j) = j$.

Fixed k. Before fixed-k retrieval, each modality's gallery is reduced to its unique representatives. The query's duplicate group is excluded by removing $r^{\mathbf{m}}(i)$, and the neighborhood is

$$\widetilde{\mathcal{N}}_k^{\mathbf{m}}(i) = \operatorname{argtopk}_{r \in \mathcal{R}^{\mathbf{m}} \setminus \{r^{\mathbf{m}}(i)\}} \mathbf{m}_i^\top \mathbf{m}_r.$$

This prevents repeated copies of the same feature from occupying several of the k positions. In the modality whose rows are already distinct, the gallery is unchanged. Alignment is then computed using the standard fixed-k neighborhood-overlap formula. This is the procedure used in the bijection experiments of Section 3.2.4.

Variable k. For variable-k, collapsing the modalities independently before thresholding would define their neighborhoods over different subsets of correspondence rows. We therefore apply the similarity threshold to the original rows and map the selected rows to their common source groups afterward:

$$\mathcal{N}_i^{\mathbf{m}} = \left\{ \pi(j) : j \in \mathcal{G} \setminus \{i\}, \quad \mathbf{m}_i^\top \mathbf{m}_j \geq \beta_i^{\mathbf{m}} \right\},$$

where $\beta_i^{\mathbf{m}}$ is the query-specific threshold defined by the variable-k protocol.

A source group is included if any of its correspondence rows lies within the query's similarity radius. Mapping rows to source groups prevents repeated copies from being counted multiple times. Only the query row itself is excluded; another correspondence from the same source remains a valid neighbor.

Because the two neighborhoods may have different cardinalities, their overlap is measured using intersection-over-union:

$$s_{\text{var}} = \frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \frac{\left|\mathcal{N}_i^{\mathbf{a}} \cap \mathcal{N}_i^{\mathbf{b}}\right|}{\left|\mathcal{N}_i^{\mathbf{a}} \cup \mathcal{N}_i^{\mathbf{b}}\right|}.$$

D What happens with non-synthetic data that is not bijective?

In the main paper (Section 3.2.4), we use the CycleReward dataset (Bahng et al., 2025) to test alignment when the bijective (one-to-one) assumption is relaxed with *synthetic* multi-modal correspondences. Here, we complement this analysis using non-synthetic many-to-many correspondences from the WIT dataset (Srinivasan et al., 2021).

D.1 Non-synthetic dataset with many-to-many correspondences

Natural duplicates in WIT. The WIT dataset naturally contains many-to-many correspondences between images and captions.

The same caption can describe many visually distinct images, and the same image is reused across Wikipedia articles with different corresponding text. Specifically, 7.1% of the captions are associated with more than one image, and 24.6% of the images have more than one caption before deduplication (see Section G.1). These naturally occurring one-to-many and many-to-one correspondences provide a complementary test bed for relaxing the bijective (one-to-one) setting without relying on generated images or captions.

Within-group deduplication. Grouping by caption text (for T2I) or by image (for I2T) can include *within-group* duplicates, i.e. a caption group may contain duplicate images, and an image group may contain repeated captions. After within-group deduplication, the number of qualifying one-to-many samples decreases from 7,844 to 4,975 for T2I and from 38,254 to 24,853 for I2T (Table 2).

We construct two complementary one-to-many datasets to mirror the experimental setup in Section 3.2.4 of the paper:

T2I (text-to-images): We select all 4,975 captions that are associated with at least 5 unique images. For each caption, we take 5 images, yielding a flat dataset of 24,875 image-text pairs.

I2T (image-to-texts): We identify 24,853 unique images that are paired with at least 5 distinct captions (by exact string matching). To match the T2I dataset size, we randomly subsample 4,975 images. For each image, we take 5 captions, again yielding 24,875 samples.

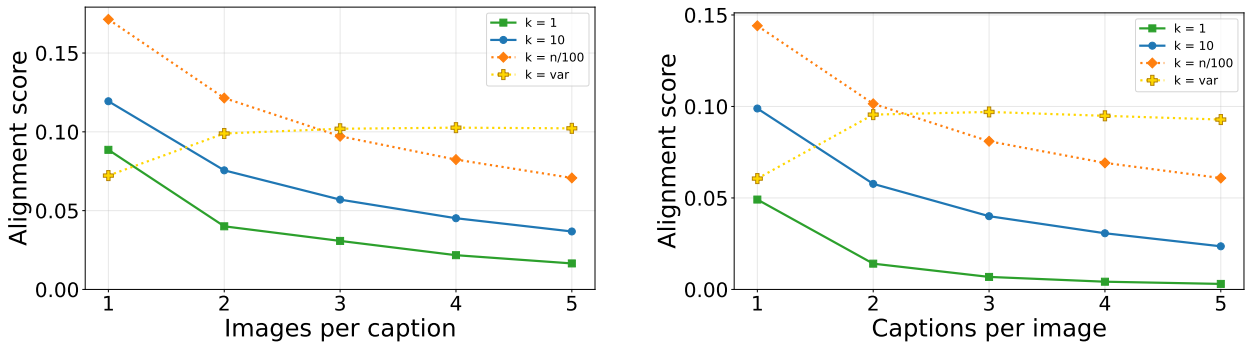
Table 2: Natural duplicates in the WIT dataset after within-group deduplication.

	T2I	I2T
Unique elements	3.2 M	2.5 M
Appearing >1 time	7.1%	24.6%
Groups with ≥5 corresp.		
Before dedup	7,844	38,254
After dedup	4,975	24,853

D.2 Mutual kNN also decreases on non-synthetic data when the bijective assumption is relaxed

We evaluate alignment between DINOv2-base (Oquab et al., 2024) and OpenLlama-3b (Geng & Liu, 2023) on the WIT-based T2I and I2T datasets. Fig. S11 shows mutual kNN alignment for k=1 and k=10 as the number of images per caption and vice versa increases from 1 to 5.

In both directions, alignment decreases as bijectivity is relaxed. This is consistent with the results on the CycleReward dataset in the main paper (Fig. 9) and reinforces the conclusion that mutual kNN alignment is sensitive to the bijective assumption. When multiple valid correspondences exist for a query, the two modalities are less likely to agree on the same nearest neighbor, even if each individually retrieves a good match. **This confirms that the observed drop in alignment for the non-bijective setting is not an artifact of *synthetic data*.**



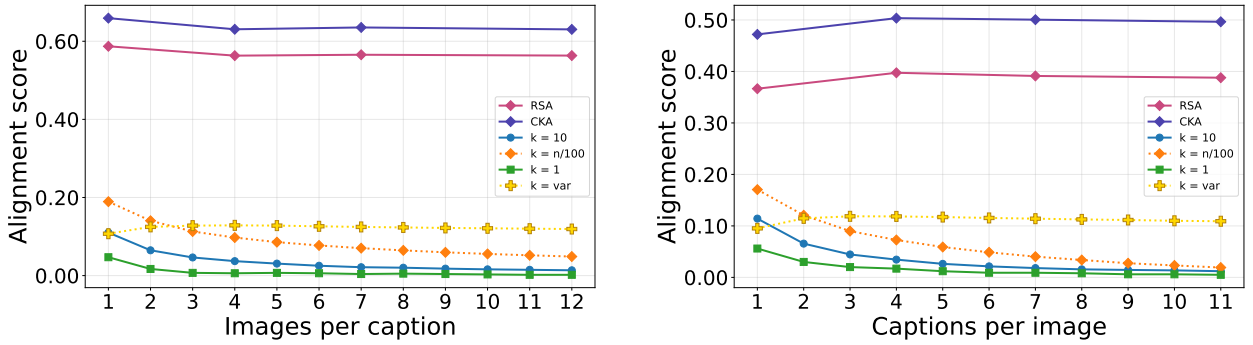
(a) T2I: increasing the number of unique images per caption from 1 (bijective) to 5.

(b) I2T: increasing the number of unique captions per image from 1 (bijective) to 5.

Figure S11: Effect of relaxing the bijective assumption on mutual kNN alignment using many-to-many correspondences in the WIT data (I2T and T2I subset). *Mutual kNN alignment on non-synthetic data drops consistently as we increase the number of images per caption and vice versa. This confirms that the pattern observed on CycleReward is not an artifact of synthetic data.*

E Is the many-to-many drop a property of the mutual kNN metric?

In Section 3.2.4, we observed that mutual kNN alignment decreases consistently as the one-to-one assumption is relaxed, and noted that the metric cannot distinguish between genuine misalignment and many-to-many correspondence. Here, we separate the two. We repeat the CycleReward densification experiments with two global metrics, RSA and linear CKA (Section C), on the identical feature pairs (DINOv2-base and OpenLlama-3b). If the drop reflected genuinely weaker alignment, i.e. a change in how the two models organize the densified data relative to one another, global measures of the pairwise structure should register it. If it is instead a property of the neighborhood-based reading, they should stay flat.



(a) T2I: increasing the number of images per caption.

(b) I2T: increasing the number of captions per image.

Figure S12: Effect of relaxing the bijective assumption on local and global metrics, for DINOv2-base and OpenLlama-3b on the CycleReward dataset (Bahng et al., 2025), using the same densified sets as Fig. 9. RSA and linear CKA remain stable as correspondences per sample are added, while mutual kNN on the identical feature pairs drops by an order of magnitude (Fig. 9).

The global metrics stay flat (Fig. S12). While mutual kNN drops by an order of magnitude as correspondences are added (k=10: 0.119 $\rightarrow$ 0.012 for T2I and 0.110 $\rightarrow$ 0.008 for I2T), CKA remains stable across the full densification range ($[0.63, 0.66]$ for T2I, $[0.47, 0.50]$ for I2T), and RSA likewise stays within a narrow band ($[0.56, 0.59]$ and $[0.37, 0.40]$, respectively) as the number of correspondences per sample grows. Adding valid correspondences thus does not change the cross-modal agreement on pairwise structure. What changes is only the composition of the top-k neighborhoods: as Fig. 8 illustrates, the densified modality fills its neighborhoods with siblings of the same source sample while the sparse modality retrieves items from other sources, and mutual kNN counts this disagreement as misalignment.

This mirrors the density control on the ImageNet validation set (Fig. 11). There, growing the gallery from 1 to 49 images per class left RSA, CKA, and the density-matched mutual kNN variants (k=n/100, variable-k) stable while fixed-k mutual kNN decayed. Consequently, the global metrics read the geometry the models impose rather than being a gallery-dependent measure, and fixed-k mutual kNN is the density-sensitive statistic. The bijection experiments show the same dissociation along a different axis of densification. Whether neighborhoods are densified by adding more samples (Section 3.2.1) or by adding more valid correspondences per sample, the representational geometry that the two models share appears to be unchanged, and only the neighborhood-based structures changes. We argue that both metric families are useful, as they measure different perspectives of alignment.

Relaxing bijectivity dissociates the two metric families. Mutual-kNN and the global metrics read different properties of the same representations: mutual-kNN reads top-k neighborhood overlap, which presupposes a unique correct match per query, while RSA and CKA read agreement on pairwise structure, which is defined regardless of how many valid correspondences a sample has.

F Does the alignment vs performance trend predicted by Huh et al. (2024) continue with recent LLMs?

To assess whether the alignment vs performance trend predicted by Huh et al. (2024) continues with recent language models, we evaluate 55 LLMs (see Section G.3.2 for full list) on six standard benchmarks using the LM Evaluation Harness framework (Gao et al., 2024). Huh et al. (2024) originally used three benchmarks to measure language capability: HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and (1 $-$ `bitsperbyte`) on OpenWebText (Gokaslan & Cohen, 2019). We replace OpenWebText with Wikitext (Merity et al., 2016) and extend this analysis to three additional benchmarks that probe different aspects of language understanding: ARC Challenge (Clark et al., 2018), MMLU (Hendrycks et al., 2021), and LogiQA2 (Liu et al., 2023).

F.1 Benchmarks and metrics.

Table 3 summarizes the evaluation configuration for each benchmark used in Section 3.2.3 of the paper (we used the default configurations from Gao et al. (2024)).

Table 3: Overview of language model benchmarks used in Section 3.2.3 of the paper.

Benchmark	Capability	Few-shot	Metric
HellaSwag (Zellers et al., 2019)	Commonsense inference	0	Accuracy
Wikitext (Merity et al., 2016)	Language modeling	0	$1 -$ bits per byte
ARC Challenge Clark et al. (2018)	Science QA	4	Accuracy
GSM8K (Cobbe et al., 2021)	Math reasoning	5	Exact match
MMLU (Hendrycks et al., 2021)	General knowledge	5	Accuracy
LogiQA2 (Liu et al., 2023)	Logical reasoning	5	Accuracy

F.2 Does the alignment vs performance trend hold?

For each benchmark and each DINOv2 variant, we fit a linear regression on the *base models* used in Huh et al. (2024), predicting mutual kNN alignment a from benchmark performance p. We then evaluate how well this trend describes two populations:

R^2 **(Huh *et al.*):** The standard coefficient of determination on the data is used to fit the regression, i.e. $R^2(\text{Huh } et\ al.) = r^2$, where r is the Pearson correlation between mutual kNN alignment and language modelling benchmark score across the 19 *base models* from Huh et al. (2024).

R^2 **(new):** We apply the line fitted on the *base models* to the 36 recent models and compute the generalized R^2:

$$R^2(\text{new}) = 1 - \frac{\sum_{i \in \mathcal{M}_{\text{new}}} (a_i - \hat{a}_i)^2}{\sum_{i \in \mathcal{M}_{\text{new}}} (a_i - \bar{a}_{\text{new}})^2},$$

where $\hat{a}_i$ are the linear regression alignment predictions based on language performance p_i, a_i is the alignment score for the i-th model and $\bar{a}_{\text{new}}$ is the mean alignment of the new models. When $R^2(\text{new}) > 0$, the relation between alignment and language performance predicted in Huh *et al.* extrapolates. When $R^2(\text{new}) < 0$, the regression line is a worse predictor than simply predicting the average $\bar{a}_{\text{new}}$. R^2_{avg} values are reported in Table 4 and Figs. S13 and S14.

The results reveal a split across language modelling benchmarks. For HellaSwag and Wikitext, the relation between alignment and language performance observed by Huh *et al.* partially extends to recent models: the R^2_{avg} on new models remains positive (0.297 and 0.489, respectively), indicating that stronger language models according to these benchmarks have higher mutual kNN alignment with DINOv2. Both benchmarks primarily measure next-token prediction quality and commonsense language understanding, which are closely related to the pretraining objective of autoregressive LLMs.

In contrast, for the four benchmarks that probe more specialized reasoning abilities: ARC (science QA), GSM8K (arithmetic), MMLU (general knowledge), and LogiQA2 (logical reasoning), the relation between alignment and language performance predicted from the *base models* (Huh et al., 2024) does not appear to hold for this set of recent models.

Specifically, the R^2_{avg} on new models is consistently negative, ranging from -0.575 (ARC) to -1.753 (GSM8K). This means that the linear fit from the *base models* from Huh et al. (2024) is a worse predictor of alignment for recent models than simply predicting the mean. In Figs. S13 and S14, we observe that recent models that are stronger than the best *base model* (Meta-Llama-3-70B) do not show higher mutual kNN alignment with DINOv2 features. Instead, their alignment scores seem to saturate or decrease.

We note that the 36 added (new) models are heterogeneous. They include new models trained on next-token prediction (pre-training), instruction-tuned models, and reasoning-distilled models (e.g. DeepSeek-R1-Distill). We treat them as a single population to test whether the trend extrapolates to recent LLMs.

Table 4: Average R^2 of the linear regression (fitted on the 19 models from Huh et al. (2024)) evaluated on the models themselves and on the 36 recent models, across all four DINOv2 variants. Positive R^2_{avg}(new) indicates that the trend from the *base models* is a good predictor for the new models. Negative values indicate that the regression line is a worse predictor than the mean. The last two columns show R^2 computed on 25 recent *base models* and 20 recent *post-trained models*.

Benchmark	R^2_{avg} **(Huh *et al.*)**	R^2_{avg} **(new)**	R^2_{avg} **(base)**	R^2_{avg} **(post-trained)**
HellaSwag	0.752	0.297	−0.231	−1.113
Wikitext	0.729	0.489	−0.266	−0.479
ARC	0.702	−0.575	−1.339	−3.611
GSM8K	0.336	−1.753	−5.110	−11.011
MMLU	0.430	−0.662	−1.801	−4.551
LogiQA2	0.431	−1.414	−3.160	−12.539

The above results support and extend the finding from Section 3.2.3 of the main paper. **The relationship between alignment and language performance from Huh et al. (2024) holds for core language modelling benchmarks, but does not seem to generalize to reasoning benchmarks.**

F.3 Is the failure to extrapolate an artifact of post-training?

We note that the negative R^2_{avg} of Section F.2 might be attributed to post-training rather than to capability. Instruction-tuned and reasoning-distilled models spread mostly at the end of the reasoning benchmarks Fig. S14, and their representations are shaped by objectives such as preference optimization and distillation from a teacher's reasoning traces. That differs from models used to predict the trend (Huh et al., 2024), although one must note that Huh et al. (2024) uses the bloomz family, which is also post-trained. Nevertheless, if post-training moved representations away from vision, the saturation would be a result of the post-training objectives rather than a breakdown of the capability–alignment relation. To separate the two, we re-evaluate the generalized R^2 on the recent models split by training regime, *base-only* against *post-trained*, holding the regression line fixed to the Huh et al. (2024) base-model fit, Table 4. For these experiments, we extended the model set from Huh et al. (2024) with 25 base-only models and 20 post-trained models.

Both subsets are negative on every benchmark: the *base-only* recent models already fail to extrapolate. The failure is mildest on the core language-modelling benchmarks (WikiText, HellaSwag), where R^2_{avg} stays close to zero, and largest on the reasoning and math benchmarks (ARC, MMLU, GSM8K, LogiQA2), where the *base-only* R^2_{avg} ranges from −1.34 to −5.11 and the *post-trained* from −3.61 to −12.54. [1]

Recent *base* models that surpass the strongest base model in Huh et al. (2024) do not exhibit higher mutual k-NN alignment with DINOv2 features either. Moreover, the gap between the two subsets has no consistent sign. Post-trained and base models are both more negative on ARC-C, MMLU, GSM8K, and LogiQA2, so the divergence does not track post-training in a systematic way. **The failure to extrapolate is therefore not an artifact of post-training: recent *base* models saturate against DINOv2 features on their own.**

F.4 Is the failure to extrapolate an artifact of the mutual kNN metric?

The trend analyses in Sections F.2 and F.3 measure alignment with mutual kNN, a local metric that reads top-k neighborhoods. One might therefore attribute the failure to extrapolate to the metric rather than to the representations. Gröger et al. (2026) report that local and global similarity measures can behave differently under calibration, and our granularity analysis (Section 3.2.1) shows that mutual kNN with k=10 at n=1024 probes the coarse regime. If the saturation were a property of how mutual kNN reads neighborhoods, global measures of representational similarity should still continue the trend. To separate the metric from the finding,

[1] R^2_{avg}(base) and R^2_{avg}(post-trained) values differ from R^2_{avg}(new) in Table 4 because the "new" subset contains a mix of base and post-trained, splitting the population changes R^2_{avg} values.

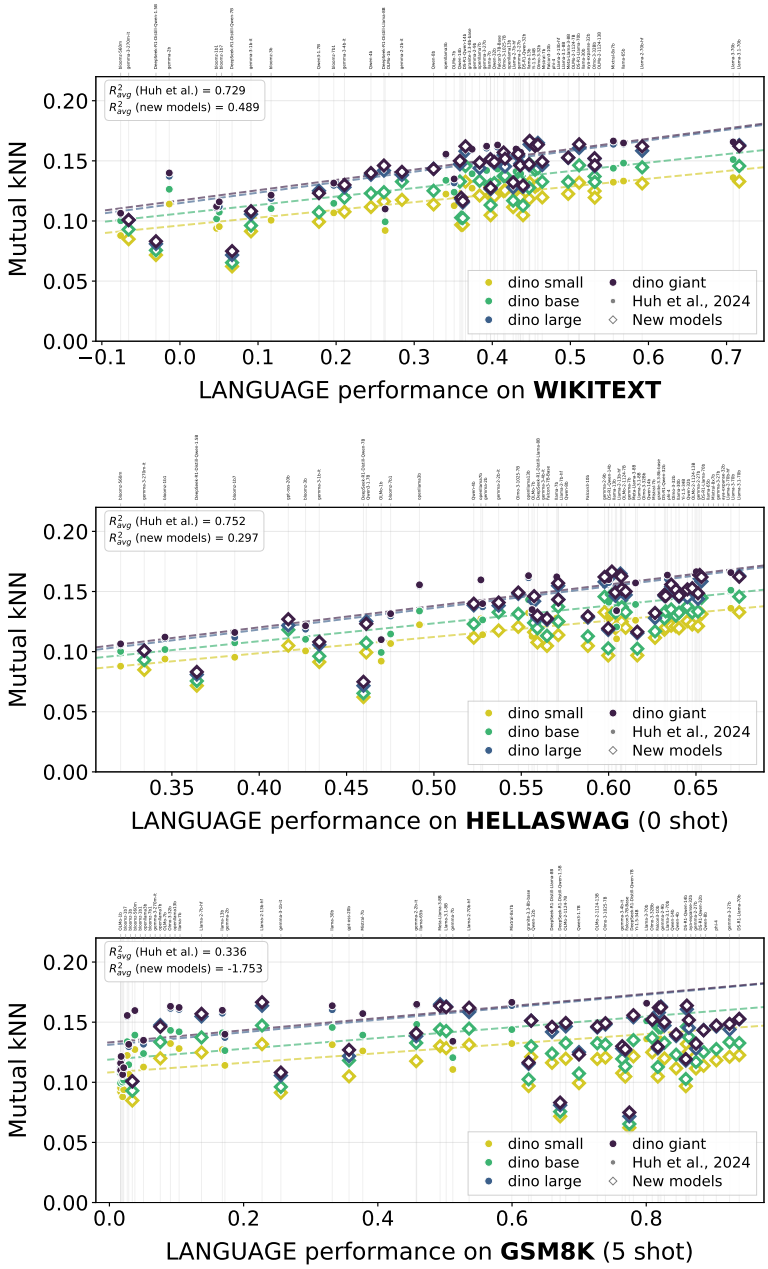


Figure S13: Mutual kNN alignment vs. language benchmark performance for 55 LLMs across four DINOv2 variants, on WikiText, HellaSwag, and GSM8K. Dashed lines show the linear trend fit to the 19 *base models* from Huh et al. (2024). For WikiText and HellaSwag (top two plots), recent models roughly follow the trend. For GSM8K (bottom plot), the trend is not followed.

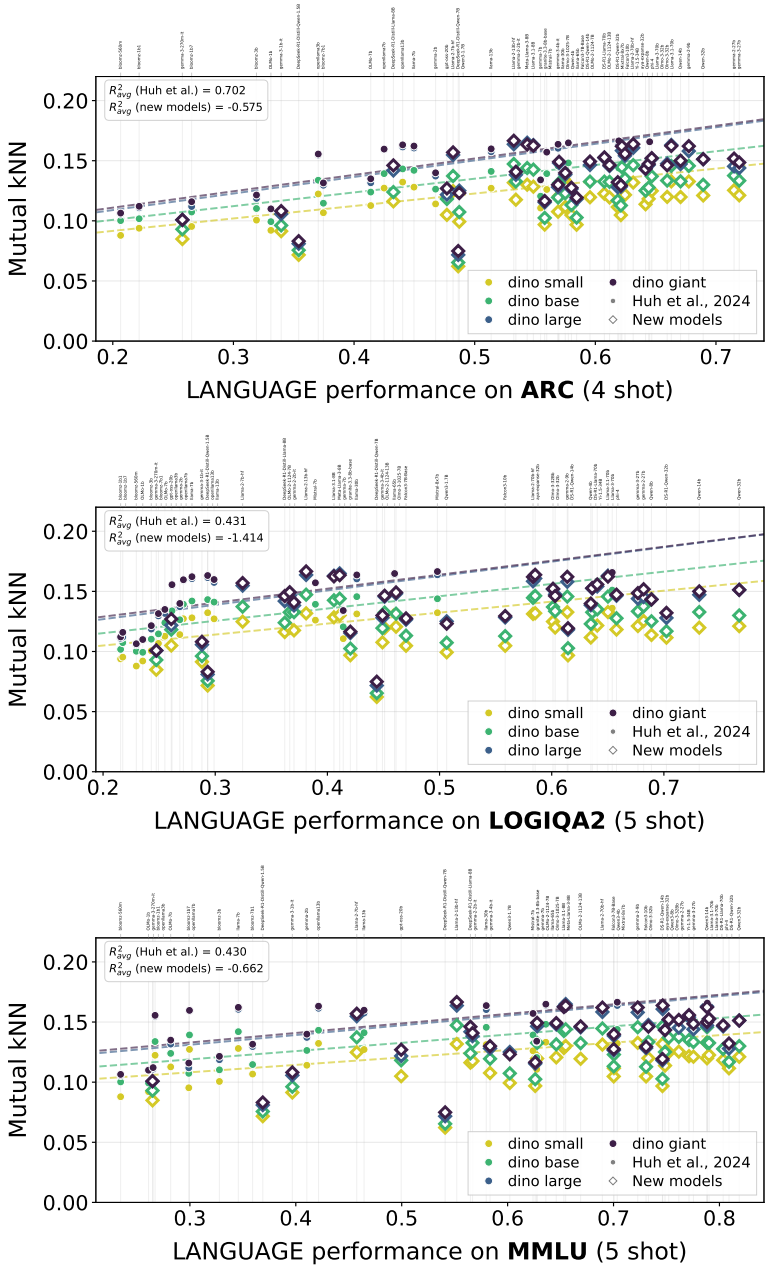


Figure S14: Mutual kNN alignment vs. language benchmark performance for 55 LLMs across four DINOv2 variants, on ARC, LogiQA2, and MMLU. As with GSM8K, the alignment-performance trend from Huh et al. (2024) does not extrapolate to recent models on any of these reasoning benchmarks. Stronger models do not appear to show higher mutual kNN alignment with DINOv2 features.

Table 5: Average R^2 of the linear regression (fitted on the 19 *base models* from Huh et al. (2024)) evaluated on the models themselves and on the recent models, for calibrated global RSA and calibrated linear CKA, averaged across the four DINOv2 variants. As in Table 4, positive R^2_{avg}(new) indicates that the trend fitted on the *base models* extrapolates to the recent models. Negative values indicate that the regression line is a worse predictor than the mean. In contrast to mutual kNN (Table 4), R^2_{avg}(new) is negative on all six benchmarks for both global metrics, including the core language-modelling benchmarks.

	RSA		**CKA**	
Benchmark	R^2_{avg} **(Huh *et al.*)**	R^2_{avg} **(new)**	R^2_{avg} **(Huh *et al.*)**	R^2_{avg} **(new)**
HellaSwag	0.602	−0.264	0.208	−0.416
Wikitext	0.685	−0.073	0.315	−0.313
ARC	0.568	−0.628	0.179	−0.754
GSM8K	0.236	−1.426	0.052	−1.218
MMLU	0.262	−0.621	0.047	−0.487
LogiQA2	0.329	−1.112	0.092	−1.280

we repeat the trend check with two global measures: RSA and linear CKA (Section C), both calibrated against the pairing-shuffle null of Gröger et al. (2026). The protocol is otherwise unchanged from Section F.2: for each benchmark and each DINOv2 variant, we fit a linear regression on the 19 *base models* from Huh et al. (2024) and evaluate the generalized R^2 on the recent models.

The saturation persists under both global metrics (Figs. S15 and S16, Table 5). On the reasoning benchmarks, R^2_{avg}(new) is consistently negative, ranging from −0.621 (MMLU) to −1.426 (GSM8K) for RSA and from −0.487 (MMLU) to −1.280 (LogiQA2) for CKA. The linear fit from the *base models* is again a worse predictor of alignment for recent models than simply predicting their mean. Notably, the global metrics fail to extrapolate even on the core language-modelling benchmarks, where mutual kNN partially extended the trend (R^2_{avg}(new) of 0.30 for HellaSwag and 0.49 for Wikitext, Table 4): under RSA and CKA, HellaSwag and Wikitext drop to −0.264/−0.073 and −0.416/−0.313, respectively. As a nonparametric check that does not presuppose a linear relationship, we additionally fit LOWESS curves over all models (bottom rows). The smoothers flatten over the capability range covered by the recent models on every benchmark, for both metrics and all four DINOv2 variants, confirming that the plateau is not an artifact of the linear functional form. The LOWESS estimation also makes it easier to see that there is no growing trend across the majority of benchmarks.

We note that for CKA the trend on the *base models* was weak to begin with (R^2_{avg}(Huh *et al.*) between 0.047 and 0.315), consistent with Huh et al. (2024), who found CKA to reveal only a "weak trend of alignment between models" and introduced mutual kNN for this reason. The CKA result should therefore be read not as a strong trend breaking down, but as no capability trend emerging under CKA for recent models either. For RSA, the *base-model* trend is substantial on the benchmarks used by Huh et al. (2024) (R^2_{avg}(Huh *et al.*) of 0.602 on HellaSwag and 0.685 on Wikitext), and it is this trend that the recent models do not continue.

The failure to extrapolate is not an artifact of the mutual kNN metric. Calibrated global RSA and CKA show the same saturation for recent models on all six benchmarks, and, unlike mutual kNN, they fail to extrapolate even on the core language-modelling benchmarks.

G Experimental setup

In Section G.1, we provide additional details about the deduplication pipeline for the WIT-1M and LAION-15M datasets. We then describe the captioning pipeline used for WIT-1M-recap and for the ImageNet validation set in Section G.2.

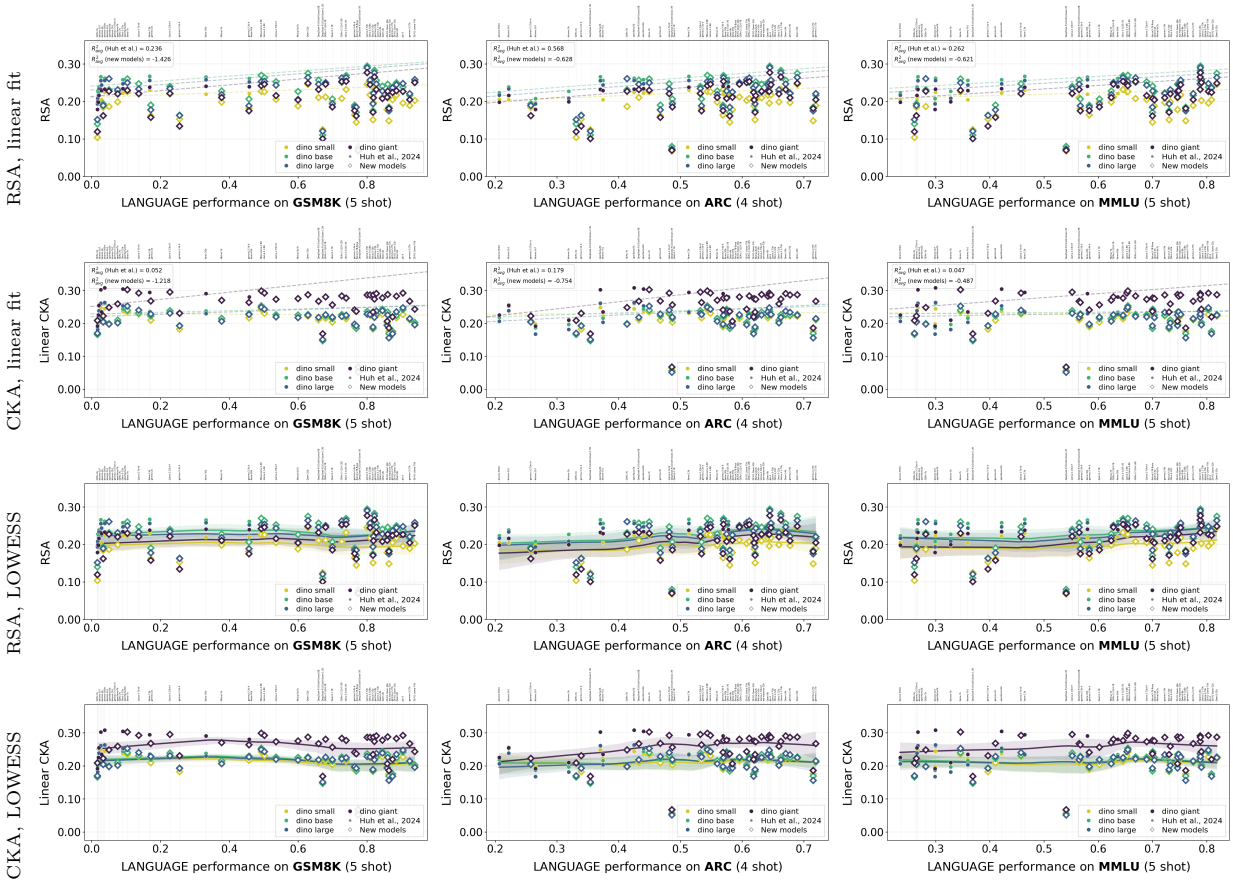


Figure S15: Calibrated global RSA (rows 1, 3) and calibrated linear CKA (rows 2, 4) between LLMs and four DINOv2 variants on WIT-1024, plotted against language performance on the reasoning benchmarks GSM8K, ARC, and MMLU (columns). The top two rows show linear fits to the 19 *base models* from Huh et al. (2024) (circles; dashed lines), evaluated on the recent models (diamonds). R^2_{avg} values are averaged over the four DINOv2 variants. The bottom two rows show LOWESS fits over all 55 models with bootstrap confidence bands. *As with mutual kNN (Figs. S13 and S14), recent models do not continue the extrapolated trend under either global metric: $R^2_{avg}(new)$ is negative throughout, and the LOWESS fits flatten over the capability range covered by the recent models.*

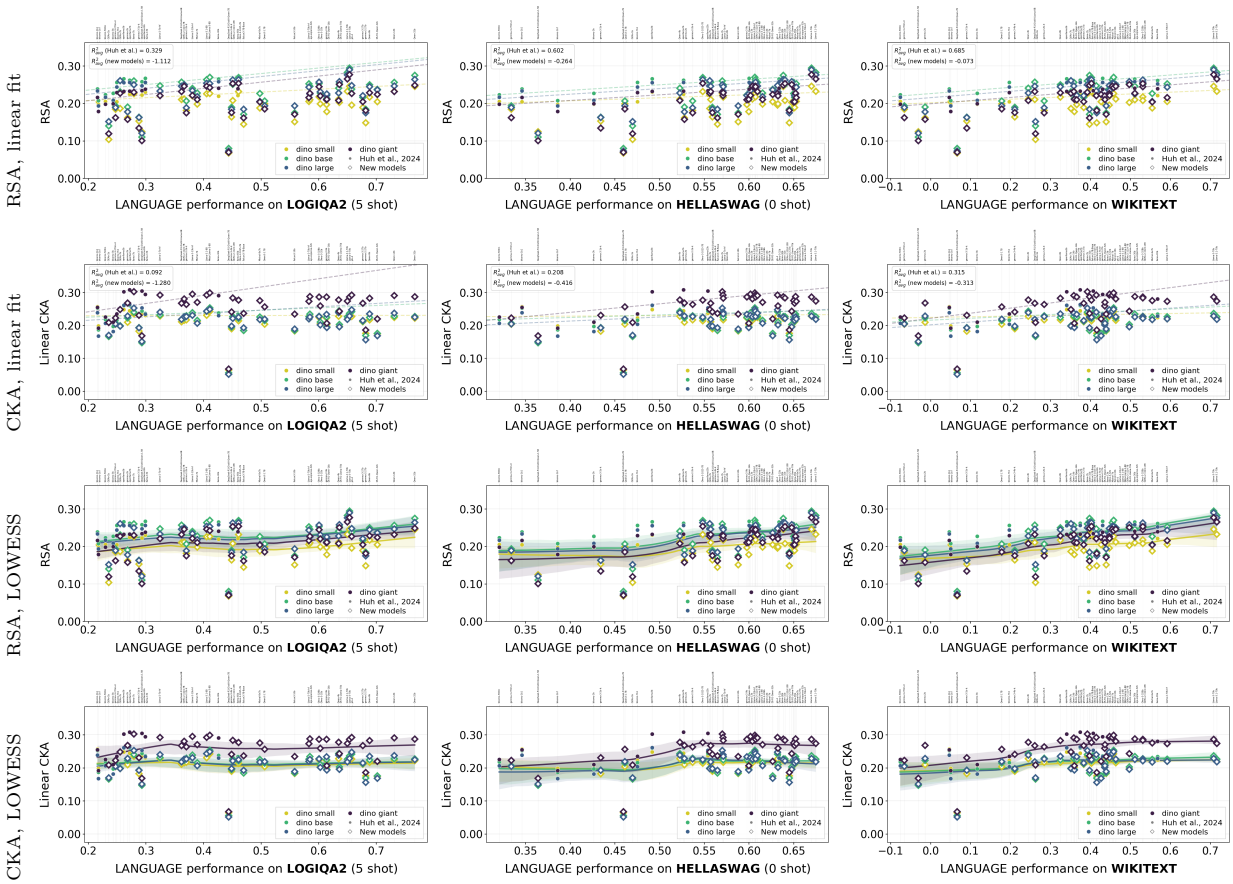


Figure S16: Calibrated global RSA (rows 1, 3) and calibrated linear CKA (rows 2, 4) between LLMs and four DINOv2 variants on WIT-1024, against language performance on LogiQA2 and the core language-modelling benchmarks HellaSwag and Wikitext (columns); layout as in Fig. S15. *Unlike mutual kNN, for which HellaSwag and Wikitext partially extended the trend to recent models ($R^2_{avg}(new)$ of* 0.30 *and* 0.49*, Table 4), the global metrics fail to extrapolate on the core language-modelling benchmarks as well.*

Table 6: Deduplication statistics for the WIT-1M and LAION-15M gallery pools. Image duplicates are detected via perceptual hashing (pHash) with Hamming distance ≤ 2. Caption duplicates are detected by exact string matching. WIT-1M and LAION-15M are 1M and 15M image-caption pairs randomly sampled from the remaining final pool.

	WIT-1M	**LAION-15M**
Raw pool	3,582,610	20,000,000
Image deduplication		
Duplicates (with WIT-1024)	2,847	53
Duplicates (within gallery)	2,164,343	3,371,128
Pool after image deduplication	2,486,852	17,941,016
Caption deduplication		
Duplicates (with WIT-1024)	52	14
Duplicates (within gallery)	97,654	642,895
Final pool size	**2,389,146**	**17,298,107**

G.1 WIT-1M and LAION-15M datasets

G.1.1 Image deduplication.

We deduplicate the gallery pools from WIT (Srinivasan et al., 2021) and LAION400M (Schuhmann et al., 2021) at the image level using perceptual hashing (Zauner, 2010). For each image, a 64-bit hash $\mathbf{h}_i \in \{0,1\}^{64}$ is computed. For this, we first convert the image to grayscale, resize it to 32×32, apply a 2D Discrete Cosine Transform, and threshold the top-left 8×8 low-frequency coefficients against their median. This produces a binary fingerprint that is robust to minor changes, e.g. due to recompression. We consider images i and j duplicates if their Hamming distance satisfies

$$d_{\mathrm{H}}(\mathbf{h}_i, \mathbf{h}_j) \leq 2,$$

measuring the number of bit positions at which two hashes differ:

$$d_{\mathrm{H}}(\mathbf{h}_i, \mathbf{h}_j) = \sum_{b=1}^{64} \mathbf{1}[h_{i,b} \neq h_{j,b}].$$

We perform deduplication of the gallery against the WIT-1024 images, and within the gallery by keeping the first occurrence in the case of image duplicates.

G.1.2 Caption deduplication.

In addition to image deduplication, we do a text deduplication pass to remove gallery samples with captions identical to another gallery sample or to a WIT-1024 query caption. Duplicate captions are undesirable because they allow trivial text-based query-gallery matching, inflating retrieval scores regardless of visual representations.

We use exact string matching and remove any gallery samples that match WIT-1024 captions. Among the gallery samples, we discard duplicates and keep only the first occurrence of a duplicate caption-sample.

G.1.3 WIT-1M.

We obtain a raw pool of 3,582,610 samples from the English-text WIT dataset (Srinivasan et al., 2021). To construct the English-only subset of the dataset, we used a subset of Singh et al. (2022). Since Singh et al. only contain the image URLs, we retrieved the corresponding images from Srinivasan et al. (2021). The raw pool undergoes our deduplication pipeline, resulting in 2,389,146 samples. We randomly sampled 1 million samples for the WIT-1M dataset. Deduplication statistics are provided in Table 6 and Table 7. In WIT, 31,845 captions appear repeatedly, accounting for 129,499 samples (5.21%). The most frequent captions are `coat of arms` (2716×) and `Town hall` (2361×).

Table 7: Distribution of caption duplicates in the WIT and LAION galleries after image deduplication. Note that the "unique captions" include some captions that are removed as query matches for the final pool.

	WIT		LAION	
Copies per caption	Unique captions	Total samples	Unique captions	Total samples
1	2,357,353	2,357,353	16,897,221	16,897,221
2	22,799	45,598	316,109	632,218
3	3,807	11,421	48,864	146,592
4	1,529	6,116	15,422	61,688
5	787	3,935	6,932	34,660
6–10	1,520	11,431	9,596	69,614
11–100	1,283	29,765	3,838	76,200
101–1,000	118	27,164	103	12,376
>1,000	2	5,077	4	14,588
Total	2,389,198	2,486,852	17,298,111	17,941,016

G.1.4 LAION-15M.

We randomly sample 20M samples from the LAION-400M dataset (Schuhmann et al., 2021) which consists of English image-text pairs. We randomly sample 20M samples as our raw pool which undergoes our deduplication pipeline. Finally, we randomly sample 15M from the final pool after deduplication, resulting in our LAION-15M data pool. Deduplication statistics are provided in Table 6 and Table 7. In LAION, 400,890 unique captions appear repeatedly, accounting for 1,043,795 samples (5.82%). The most frequent captions are `Patent Drawing` (10027×) and `Throw Pillow` (3246×).

G.2 Captioning pipeline for the ImageNet validation set and WIT-1M-recap

We used gemini-3-flash-preview (Team et al., 2023; Pichai et al., 2025) for captioning the images in the ImageNet validation (Deng et al., 2009) set and in WIT-1M. Specifically, we used the following text prompt for each image.

```
You are a precise image description system. Describe the image in the following JSON format.
Return ONLY a valid JSON object with exactly these 7 keys. No text before or after the JSON.
{
  "one_sentence": "<exactly one sentence, strictly fewer than 15 words>",
  "short":        "<2-3 sentences, 20-40 words total>",
  "100w":         "<a paragraph, approximately 100 words>",
  "250w":         "<several paragraphs, approximately 250 words>",
  "500w":         "<detailed description, approximately 500 words>",
  "750w":         "<thorough description covering all visual details, approximately 750 words>",
  "extreme_long": "<maximally detailed description covering every visible element, texture, color,
      spatial relationship, lighting, and context. YOU MUST WRITE AT LEAST 1000 WORDS. If your draft is
      under 1000 words, keep adding more detail about textures, materials, lighting, spatial layout,
      colors, and any other visible elements until you reach at least 1000 words.
  Target 1000-1500 words.>"
}
Be factual and visual. Describe what you actually see: objects, people, animals, colors, textures,
    spatial relationships, background, lighting, and mood. Do not invent information not visible in the
    image.
```

For WIT-1M-recap, we caption the 1 million images in the WIT-1M dataset using the `500w` variant for computational efficiency. This resulted in captions for 999,971 images (29 images did not get processed by gemini-3-flash-preview) of on average 478 words.

G.3 Models and feature extraction pipeline

Our feature extraction pipeline is based on the experimental protocol from Huh et al. (2024), which we extend to include additional LLMs.

G.3.1 Vision models.

On the vision side, we use four DINOv2 (Oquab et al., 2024) variants: ViT-S/14 (384-d), ViT-B/14 (768-d), ViT-L/14 (1024-d), and ViT-G/14 (1536-d) (Dosovitskiy et al., 2021), loaded via the `timm` (Wightman, 2019) library. For each image, we extract the CLS token representation from every transformer layer, yielding a per-sample feature tensor of shape $L \times d$, where L is the number of layers and d the feature dimensionality.

G.3.2 Language models.

We evaluate 55 large language models spanning 13 model families. The first group comprises the 19 *base models* used by Huh et al. (2024): BLOOMZ (560M–7.1B) (Muennighoff et al., 2023), OpenLlama (3B–13B) (Geng & Liu, 2023), LLaMA (7B–65B) (Touvron et al., 2023a), OLMo (1B, 7B) (Groeneveld et al., 2024), Gemma (2B, 7B) (Gemma Team, 2024a), Mistral-7B (Jiang et al., 2023), Mixtral-8×7B (Jiang et al., 2024), and Meta-Llama-3-70B (Grattafiori et al., 2024).
For the trend analysis in Section F, we extend this set with 36 recent models: LLaMA-2 (7B–70B) (Touvron et al., 2023b), Llama-3/3.1 (Grattafiori et al., 2024), OLMo-2/3 (OLMo et al., 2025), Gemma-2 (2B–27B)(Gemma Team, 2024b), Gemma-3 (270M–27B) (Gemma Team, 2025), DeepSeek-R1-Distill (1.5B–70B) (DeepSeek-AI et al., 2025), Qwen3 (1.7B–32B) (Qwen Team, 2025), Falcon3 (7B, 10B) (Team, 2024), 01.AI Yi-1.5 (34B) (Young et al., 2024), IBM Granite (8B) (Research, 2025), Phi-4 (15B) (Abdin et al., 2024), Cohere Aya Expanse (32B) (Dang et al., 2024) and OpenAI GPT-OSS (20B) (OpenAI, 2025).
For each model, we extract hidden-state representations from all layers. Following Huh et al. (2024), we apply average pooling over non-padding tokens to obtain a single vector per layer.

H Additional qualitative results

We present additional qualitative results for nearest-neighbor retrieval at different gallery scales on LAION-15M and WIT-1M in Figs. S17 to S19 and Figs. S20 and S21 respectively. Here, we show further examples for cross-modal agreement (green-bordered matches) at scale when the modalities happen to select the same neighbor. Others show agreement at WIT-1024 that breaks down as the gallery densifies. In those cases, each modality individually finds a better match at scale, but they no longer agree on the same one.

Figure S17: Nearest-neighbor (k=1) examples with DINOv2 and OpenLlama across gallery scales on LAION-15M. *As the gallery densifies, each model finds closer but different matches (top example). The match at 15M (bottom right) is a near-duplicate that survived our deduplication pipeline.*

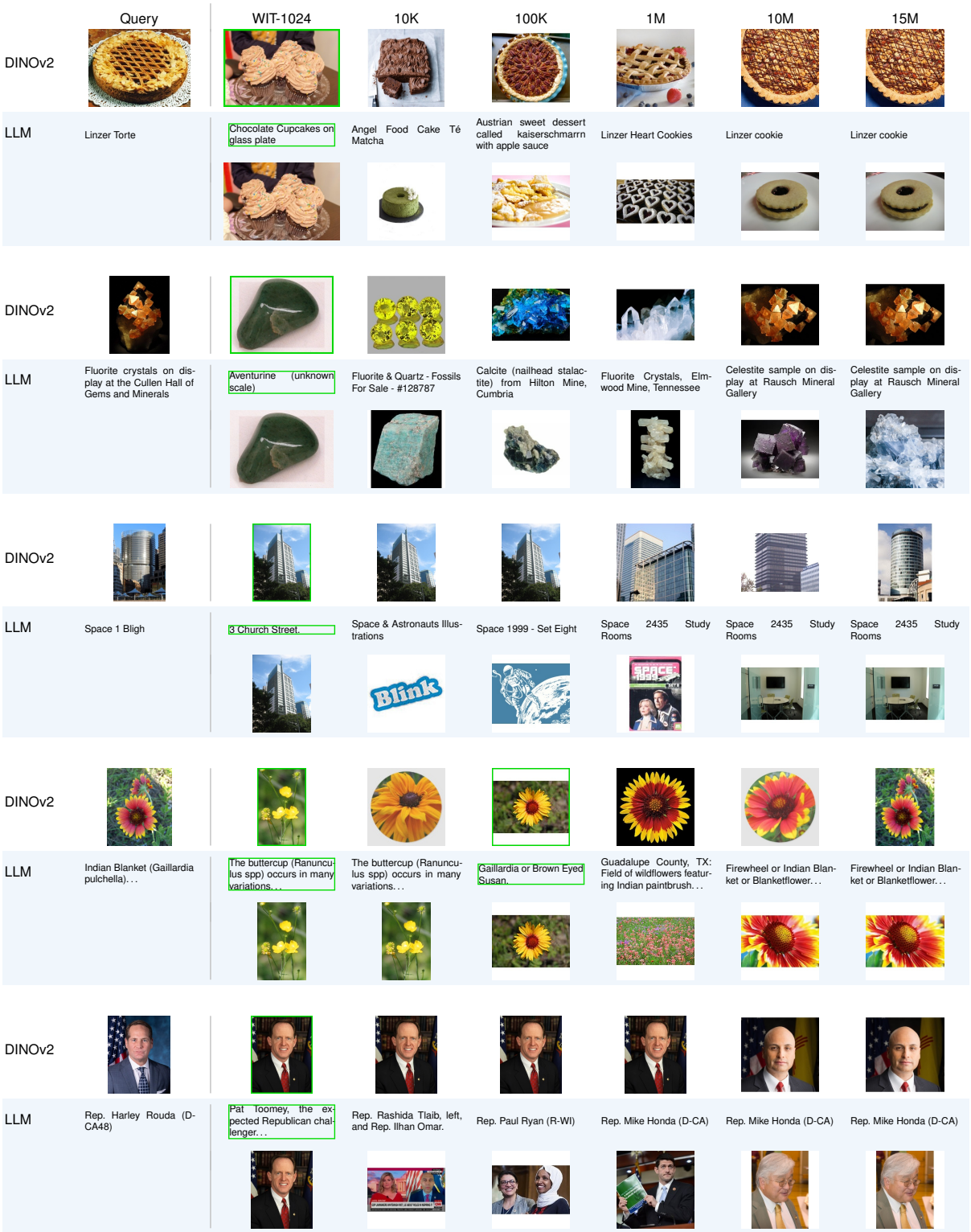


Figure S18: Additional nearest-neighbor examples with DINOv2 and OpenLlama-3b for k=1 across gallery scales on the LAION-15M dataset. Green-bordered captions and images indicate a mutual kNN match across modalities.

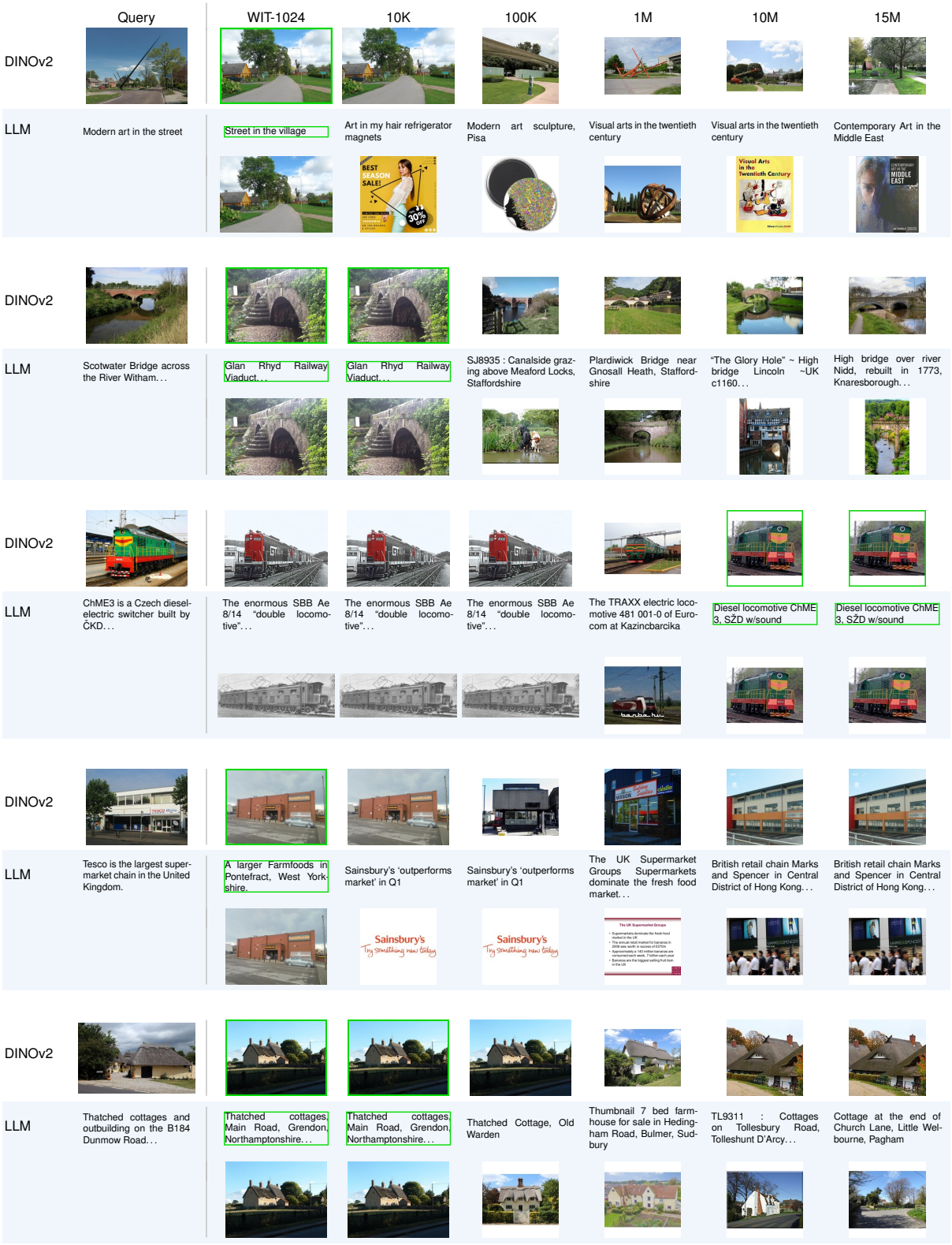


Figure S19: Additional nearest-neighbor examples with DINOv2 and OpenLlama-3b for k=1 across gallery scales on the LAION-15M dataset. Green-bordered captions and images indicate a mutual kNN match across modalities.

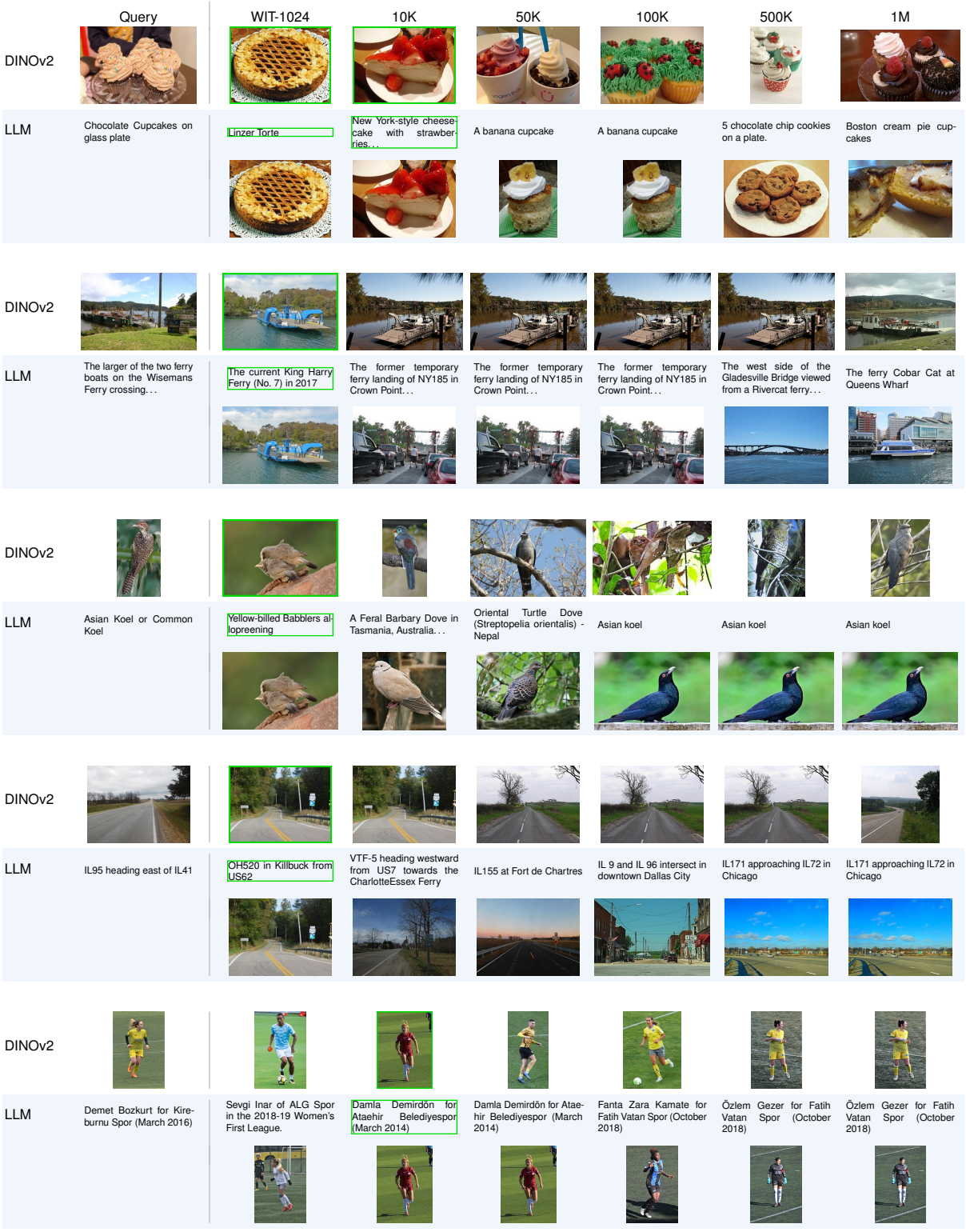


Figure S20: Additional nearest-neighbor examples with DINOv2 and OpenLlama-3b for k=1 across gallery scales on the WIT-1M dataset. For OpenLlama-3b, we show the (partial) retrieved captions along with the corresponding reference image (LLM-ref) for visualization. Green-bordered captions and images indicate a mutual kNN match across modalities.

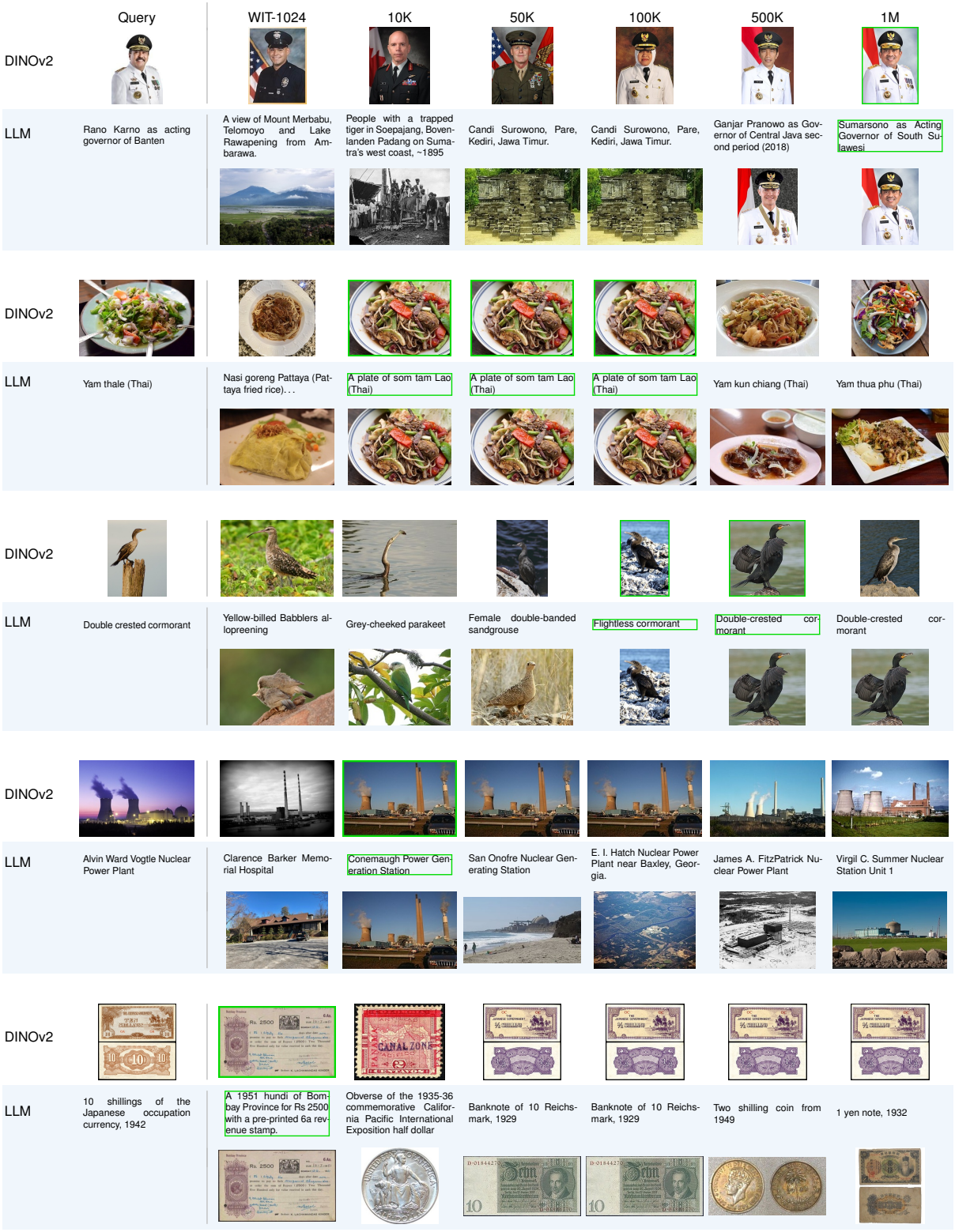


Figure S21: Additional nearest-neighbor examples with DINOv2 and OpenLlama-3b for k=1 across gallery scales on the WIT-1M dataset. Green-bordered captions and images indicate a mutual kNN match across modalities.

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